PROPERTIES OF THE GENERALIZED DIXON ELLIPTIC FUNCTIONS AND AN ALTERNATIVE FORMULATION OF RAMANUJAN'S BASE 3 ELLIPTIC FUNCTIONS

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Abstract. In this paper, we will study the properties of generalized Dixon elliptic functions, which are a set of elliptic functions that parameterize the curve $x^3 + 3\alpha xy + y^3 = 1$, first studied by A.C. Dixon and later by P.L. Robinson. We will study the functions first as a function of a complex variable z and R a complex number that is related to the nome $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. Then, we will see how we can instead study the functions as a function of z and the parameter α in a way analogous to k in Jacobi's classical elliptic functions. We will also show how they are related to the theory of Ramanujan's base 3 elliptic functions, which were studied thoroughly by Bruce C. Berndt, S. Bhargava, and Frank G. Garvan, and J.M. and P.B. Borwein, and prove several of their results using the Dixon elliptic functions. Lastly, we will derive several series for $\frac{1}{\pi}$ that were given by S. Ramanujan, H. Chan, W. Liaw, and V. Tan using properties of modular functions in this theory. The ultimate goal of this paper is to unify the theory of Dixon elliptic functions with that of Ramanujan's base 3 elliptic functions.

1. Introduction

The study of Dixon elliptic functions $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$, as done by Dixon and Robinson, starts by treating the parameter α as an abstract constant associated to the cubic curve $x^3 + 3\alpha xy + y^3 = 1$ and develops the associated elliptic function theory from this standpoint. In contrast, in the present work, α, $\mathrm{sm}(z)$, and $\mathrm{cm}(z)$ are expressed explicitly in terms of theta functions associated with the nome, and we show how this standpoint results in the equivalent formulation by Dixon and Robinson as well as an equivalent formulation of base 3 elliptic functions by Berndt, Bhargava, and Garvan.

We start by considering the following functions:

$$f(z,\tau) = \frac{\sin(\pi z)}{\sin\left(\pi z + \frac{\pi}{3}\right)} \prod_{n=1}^{\infty} \frac{1 - 2e^{2\pi i n\tau}\cos(2\pi z) + e^{4\pi i n\tau}}{1 - 2e^{2\pi i n\tau}\cos\left(2\pi z + \frac{2\pi}{3}\right) + e^{4\pi i n\tau}},$$

$$g(z,\tau) = \frac{\sin\left(\frac{\pi}{3} - \pi z\right)}{\sin\left(\frac{\pi}{3} + \pi z\right)} \prod_{n=1}^{\infty} \frac{1 - 2e^{2\pi i n\tau}\cos\left(\frac{2\pi}{3} - 2\pi z\right) + e^{4\pi i n\tau}}{1 - 2e^{2\pi i n\tau}\cos\left(\frac{2\pi}{3} + 2\pi z\right) + e^{4\pi i n\tau}}$$

for $\Im(\tau) > 0$. We can rewrite these functions in terms of Jacobi's theta functions. Let

$$\theta_1(z,\tau) = 2e^{\frac{i\pi\tau}{4}} \sin(z) \prod_{n=1}^{\infty} \left(1 - e^{2\pi i n\tau}\right)\left(1 - 2e^{2\pi i n\tau}\cos(2z) + e^{4\pi i n\tau}\right)$$

be one of Jacobi's theta functions. Then, we can see that $f(z,\tau) = \frac{\theta_1(\pi z,\tau)}{\theta_1\left(\pi z+\frac{\pi}{3},\tau\right)}$ and $g(z,\tau) = \frac{\theta_1\left(\frac{\pi}{3}-\pi z,\tau\right)}{\theta_1\left(\frac{\pi}{3}+\pi z,\tau\right)}$. Using the quasi-periodicity of the theta functions, we can see that f and g are doubly periodic, meromorphic functions. That is, they are elliptic functions. In particular they satisfy the following equations:

- $f(z+1,\tau) = f(z+3\tau,\tau) = f(z,\tau)$
- $g(z+1,\tau) = g(z+3\tau,\tau) = g(z,\tau)$
- $f\left(\frac{1}{3}-z,\tau\right) = g(z,\tau)$ and $g\left(\frac{1}{3}-z,\tau\right) = f(z,\tau)$

Moreoever, from the product definitions, we can see that $f(-z,\tau) = -\frac{f(z,\tau)}{g(z,\tau)}$ and $g(-z,\tau) = \frac{1}{g(z,\tau)}$.

We will now recast these formulations in terms of another parameter R, which will be related to τ.

Let $q = e^{2\pi i\tau} = e^{-\frac{2\pi R}{\sqrt{3}}}$, so that $\tau = \frac{iR}{\sqrt{3}}$ for $\Re(R) > 0$. As it stands, the functions f and g have periods 1 and $i\sqrt{3}R$. We will write $R = \frac{K_3'}{K_3}$, where $K_3 := \frac{\pi}{2\sqrt{3}}\prod_{n=1}^{\infty}\frac{(1-q^n)^3}{1-q^{3n}}$ and $K_3' := RK_3$. In later sections, we will see how we can write K_3 and K_3' in other ways, specifically in the form of elliptic integrals. We then define the elliptic functions $\mathrm{sm}(z) = f\left(\frac{z}{4K_3}, R\right)$ and $\mathrm{cm}(z) = g\left(\frac{z}{4K_3}, R\right)$. We suppress the dependence on R in the notation for $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$, but both functions are defined via R and z. That is,

$$\mathrm{sm}(z) := \frac{\sin\left(\frac{\pi z}{4K_3}\right)}{\sin\left(\frac{\pi z}{4K_3}+\frac{\pi}{3}\right)} \prod_{n=1}^{\infty} \frac{1 - 2e^{-\frac{2\pi nR}{\sqrt{3}}}\cos\left(\frac{\pi z}{2K_3}\right) + e^{-\frac{4\pi nR}{\sqrt{3}}}}{1 - 2e^{-\frac{2\pi nR}{\sqrt{3}}}\cos\left(\frac{\pi z}{2K_3}+\frac{2\pi}{3}\right) + e^{-\frac{4\pi nR}{\sqrt{3}}}},$$

$$\mathrm{cm}(z) := \frac{\sin\left(\frac{\pi}{3}-\frac{\pi z}{4K_3}\right)}{\sin\left(\frac{\pi}{3}+\frac{\pi z}{4K_3}\right)} \prod_{n=1}^{\infty} \frac{1 - 2e^{-\frac{2\pi nR}{\sqrt{3}}}\cos\left(\frac{2\pi}{3}-\frac{\pi z}{2K_3}\right) + e^{-\frac{4\pi nR}{\sqrt{3}}}}{1 - 2e^{-\frac{2\pi nR}{\sqrt{3}}}\cos\left(\frac{2\pi}{3}+\frac{\pi z}{2K_3}\right) + e^{-\frac{4\pi nR}{\sqrt{3}}}},$$

$$K_3 := \frac{\pi}{2\sqrt{3}} \prod_{n=1}^{\infty} \frac{\left(1 - e^{-\frac{2\pi nR}{\sqrt{3}}}\right)^3}{1 - e^{-2\sqrt{3}\pi nR}}.$$

Thus, we see that K_3 and K_3' are related to the two periods of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ and are functions of R. Moreover, the definition of K_3 may seem arbitrary, but it was chosen so that $\mathrm{sm}'(0) = 1$.

We have that $\mathrm{sm}\,(z+4K_3) = \mathrm{sm}\left(z+4i\sqrt{3}K_3'\right) = \mathrm{sm}(z)$ and $\mathrm{cm}\,(z+4K_3) = \mathrm{cm}(z+4i\sqrt{3}K_3') = \mathrm{cm}(z)$. The functions are also complements of each other, as $\mathrm{sm}\left(\frac{4K_3}{3}-z\right) = \mathrm{cm}(z)$. It can be seen that they satisfy the relations $\mathrm{sm}(-z) = -\frac{\mathrm{sm}(z)}{\mathrm{cm}(z)}$ and $\mathrm{cm}(-z) = \frac{1}{\mathrm{cm}(z)}$. The quasi-periodicity of θ_1 implies the more general shift relations $\mathrm{sm}\left(z+\frac{4inK_3'}{\sqrt{3}}\right) = \omega^n\mathrm{sm}(z)$ and $\mathrm{cm}\left(z+\frac{4inK_3'}{\sqrt{3}}\right) = \omega^{2n}\mathrm{cm}(z)$, where $\omega = e^{\frac{2\pi i}{3}}$ and n is an integer. As the naming and properties of these functions suggest, these functions seem to be equivalent to the Dixon elliptic functions, introduced by A.C. Dixon in [1]. We will see in the next section that this is indeed the case.

It should also be noted that the case $\alpha = 0$, which was studied thoroughly by P.L. Robinson in [2] and can also be found in [3] by S. Cooper, results from the case $R = e^{i\pi/3}$. Much of this paper will be devoted to studying the Dixon elliptic function in its most general form and in a way similar to that of the Jacobi elliptic functions.

2. Algebraic Relationship between $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ and Connection with Dixon Elliptic Functions

Using the known poles and zeros of the theta function θ_1, we can see that $\mathrm{sm}(z)$ has simple zeros at $z = 4mK_3 + \frac{4inK_3'}{\sqrt{3}}$ and simple poles at $z = \frac{4K_3}{3}(3m+2) + \frac{4inK_3'}{\sqrt{3}}$. The function $\mathrm{cm}(z)$ has simple zeros at $z = \frac{4K_3}{3}(3m+1) + \frac{4inK_3'}{\sqrt{3}}$ and simple poles at $z = \frac{4K_3}{3}(3m+2) + \frac{4inK_3'}{\sqrt{3}}$. Moreover, the functions have the following residues at these poles:

$$\operatorname*{Res}_{z=\frac{4K_3}{3}(3m+2)+\frac{4inK_3'}{\sqrt{3}}} \mathrm{sm}(z) = -\omega^n,$$

$$\operatorname*{Res}_{z=\frac{4K_3}{3}(3m+2)+\frac{4inK_3'}{\sqrt{3}}} \mathrm{cm}(z) = \omega^{2n}.$$

Identification of the poles and zeros of these two functions, leads to the following theorem:

Theorem 2.1. *There exists a function α, only dependent on R, such that*

$$\mathrm{sm}^3(z) + 3\alpha\mathrm{sm}(z)\mathrm{cm}(z) + \mathrm{cm}^3(z) = 1$$

for all $z \in \mathbb{C}$.

Proof. Let R be fixed. Theorem 2.1 is equivalent to the fact that $w(z) = \frac{1-\mathrm{sm}^3(z)-\mathrm{cm}^3(z)}{3\mathrm{sm}(z)\mathrm{cm}(z)}$ is constant in z. The only place w could have poles is at the poles of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ or at the zeros of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$. We first show that w has no poles at the poles of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$. Let $p_{m,n} = \frac{4K_3}{3}(3m+2) + \frac{4inK_3'}{\sqrt{3}}$.Using the residues of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$, we can expand $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ near each pole $p_{m,n}$ to get $\mathrm{sm}(z) = -\frac{\omega^n}{z-p_{m,n}} + O(1)$,

$\mathrm{cm}(z) = \frac{\omega^{2n}}{z-p_{m,n}} + O(1)$, $\mathrm{sm}^3(z) = -\frac{1}{(z-p_{m,n})^3} + O\left(\frac{1}{(z-p_{m,n})^2}\right)$, and $\mathrm{cm}^3(z) = \frac{1}{(z-p_{m,n})^3} + O\left(\frac{1}{(z-p_{m,n})^2}\right)$. Thus,

$$w(z) = \frac{1 - \left(-\frac{1}{(z-p_{m,n})^3} + O\left(\frac{1}{(z-p_{m,n})^2}\right)\right) - \left(\frac{1}{(z-p_{m,n})^3} + O\left(\frac{1}{(z-p_{m,n})^2}\right)\right)}{3\left(-\frac{\omega^n}{z-p_{m,n}} + O(1)\right)\left(\frac{\omega^{2n}}{z-p_{m,n}} + O(1)\right)}$$
$$= \frac{O\left(\frac{1}{(z-p_{m,n})^2}\right)}{O\left(\frac{1}{(z-p_{m,n})^2}\right)} = O(1).$$

Therefore, $w(z)$ has no poles at any of the poles of $\mathrm{sm}(z)$ or $\mathrm{cm}(z)$. Next, we look near the zeros of $\mathrm{sm}(z)$.

We have that $\mathrm{cm}\left(4mK_3 + \frac{4inK_3'}{\sqrt{3}}\right) = \omega^{2n}\mathrm{cm}(4mK_3) = \omega^{2n}$. Let $z_{m,n} = 4mK_3 + \frac{4inK_3'}{\sqrt{3}}$, then we have that $\mathrm{cm}(z_{m,n}) = \omega^{2n}$. We expand near $z - z_{m,n}$ to get $\mathrm{sm}(z) = O(z - z_{m,n})$, $\mathrm{cm}(z) = \omega^{2n} + O(z - z_{m,n})$, $\mathrm{sm}^3(z) = O((z - z_{m,n})^3)$, and $\mathrm{cm}^3(z) = 1 + O(z - z_{m,n})$. Thus,

$$w(z) = \frac{O\left((z - z_{m,n})^3\right) + O(z - z_{m,n})}{O(z - z_{m,n}) + O\left((z - z_{m,n})^2\right)} = \frac{O(z - z_{m,n})}{O(z - z_{m,n})} = O(1).$$

Thus, $w(z)$ has no poles at any of the zeros of $\mathrm{sm}(z)$.

From the product definition of $\mathrm{sm}(z)$, it can be seen that $\mathrm{sm}\left(\frac{4K_3}{3}\right) = 1$. From this, it follows that $\mathrm{sm}\left(\frac{4K_3}{3}(3m+1) + \frac{4inK_3'}{\sqrt{3}}\right) = \omega^n$. Let $z_{m,n}' = \frac{4K_3}{3}(3m+1) + \frac{4inK_3'}{\sqrt{3}}$ which are the zeros of $\mathrm{cm}(z)$. We can expand $\mathrm{sm}(z)$, $\mathrm{cm}(z)$, $\mathrm{sm}^3(z)$, and $\mathrm{cm}^3(z)$ near $z - z_{m,n}'$. We have $\mathrm{sm}(z) = \omega^n + O(z - z_{m,n}')$, $\mathrm{cm}(z) = O(z - z_{m,n}')$, $\mathrm{sm}^3(z) = 1 + O(z - z_{m,n}')$, and $\mathrm{cm}^3(z) = O((z - z_{m,n}')^3)$. Thus,

$$w(z) = \frac{1 - \left(1 + O\left(z - z_{m,n}'\right)\right) + O\left(\left(z - z_{m,n}'\right)^3\right)}{3\left(\omega^n + O\left(z - z_{m,n}'\right)\right)O\left(z - z_{m,n}'\right)} = \frac{O\left(z - z_{m,n}'\right)}{O\left(z - z_{m,n}'\right)} = O(1).$$

Since $w(z)$ is a rational function of elliptic functions, we have that $w(z)$ is an entire elliptic function. By Liouville's theorem, $w(z)$ must be constant, which proves the theorem. We call this constant α, which is purely a function of R. □

Theorem 2.2. *Let R be a complex number with $\Re(R) > 0$ and $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. Then,*

$$\alpha = \frac{2\theta_1'(\frac{\pi}{3}, \frac{iR}{\sqrt{3}})}{\theta_1'\left(0, \frac{iR}{\sqrt{3}}\right)} = 1 + 9q\prod_{n=1}^{\infty}\frac{\left(1 - q^{9n}\right)^3}{\left(1 - q^n\right)^3}$$

where $\theta_1'(z, \tau) = \frac{\partial}{\partial z}\theta_1(z, \tau)$.

Proof. The function α is defined to be the function $\frac{1-\mathrm{sm}^3(z)-\mathrm{cm}^3(z)}{3\mathrm{sm}(z)\mathrm{cm}(z)}$, which is a constant in z by theorem 2.1. Thus,

$$\alpha = \lim_{z\to 0}\frac{1-\mathrm{sm}^3(z)-\mathrm{cm}^3(z)}{3\mathrm{sm}(z)\mathrm{cm}(z)} = \lim_{z\to 0}\frac{1-\mathrm{cm}(z)}{\mathrm{sm}(z)} = \lim_{z\to 0}\frac{\theta_1\left(\frac{\pi}{3}+\pi z,\ \frac{iR}{\sqrt{3}}\right)-\theta_1\left(\frac{\pi}{3}-\pi z,\ \frac{iR}{\sqrt{3}}\right)}{\theta_1\left(\pi z,\ \frac{iR}{\sqrt{3}}\right)}.$$

we can then use L'Hospital's rule to get that $\alpha = \frac{2\theta_1'(\frac{\pi}{3},\frac{iR}{\sqrt{3}})}{\theta_1'\left(0,\frac{iR}{\sqrt{3}}\right)}$. To prove the second equality, we use the series representation of $\theta_1'(z,\tau)$ to get

$$\alpha = \frac{2\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{\pi\left(n+\frac{1}{2}\right)^2R}{\sqrt{3}}}\cos\left(\frac{(2n+1)\pi}{3}\right)}{\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{\pi\left(n+\frac{1}{2}\right)^2R}{\sqrt{3}}}}$$

$$= \frac{9\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{9\pi\left(n+\frac{1}{2}\right)^2R}{\sqrt{3}}}+\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{\pi\left(n+\frac{1}{2}\right)^2R}{\sqrt{3}}}}{\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{\pi\left(n+\frac{1}{2}\right)^2R}{\sqrt{3}}}}$$

$$= 1+9e^{-\frac{2\pi R}{\sqrt{3}}}\frac{\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{18\pi R}{\sqrt{3}}\frac{n(n+1)}{2}}}{\sum_{n=0}^{\infty}(-1)^n(2n+1)\,e^{-\frac{2\pi R}{\sqrt{3}}\frac{n(n+1)}{2}}}.$$

We then use Jacobi's identity, which states $\sum_{n=0}^{\infty}(-1)^n(2n+1)\,x^{\frac{n(n+1)}{2}} = \prod_{n=1}^{\infty}(1-x^n)^3$ to finish the proof. □

Theorem 2.3. *The functions* $\mathrm{sm}(z)$ *and* $\mathrm{cm}(z)$ *satisfy the differential equations*

$$\frac{d}{dz}\mathrm{sm}(z) = \mathrm{cm}^2(z)+\alpha\ \mathrm{sm}(z), \qquad \mathrm{sm}(0)=0,$$

$$\frac{d}{dz}\mathrm{cm}(z) = -\mathrm{sm}^2(z)-\alpha\ \mathrm{cm}(z), \qquad \mathrm{cm}(0)=1.$$

Proof. We start by proving the derivative $\mathrm{sm}(z)$. The derivative of $\mathrm{cm}(z)$ will follow by theorem 2.1. Let $H(z) = \frac{d}{dz}\mathrm{sm}(z)-\mathrm{cm}^2(z)-\alpha\mathrm{sm}(z)$. This function could only possibly have poles at the poles of $\mathrm{sm}(z)$,which are also the poles of $\mathrm{cm}(z)$. We expand near each pole $p_{m,n} = \frac{4K_3}{3}(3m+2)+\frac{4inK_3'}{\sqrt{3}}$. First, we calculate $\lim_{z\to p_{m,n}}\mathrm{cm}(z)-\frac{\omega^{2n}}{z-p_{m,n}} = \lim_{z\to 0}\mathrm{cm}(z+p_{m,n})-\frac{\omega^{2n}}{z}$, which is the constant term for the Laurent series of $\mathrm{cm}(z)$ around $p_{m,n}$. Using the periodicity and quasi-periodicity of θ_1, we have that

$$\mathrm{cm}(z+p_{m,n}) = \frac{\theta_1\left(\frac{\pi}{3}-\frac{\pi z}{4K_3}-\frac{\pi}{4K_3}p_{m,n},\ \frac{iR}{\sqrt{3}}\right)}{\theta_1\left(\frac{\pi}{3}+\frac{\pi z}{4K_3}+\frac{\pi}{4K_3}p_{m,n},\ \frac{iR}{\sqrt{3}}\right)} = \omega^{2n}\frac{\theta_1\left(\frac{\pi z}{4K_3}+\frac{\pi}{3},\ \frac{iR}{\sqrt{3}}\right)}{\theta_1\left(\frac{\pi z}{4K_3},\ \frac{iR}{\sqrt{3}}\right)} = \frac{\omega^{2n}}{\mathrm{sm}\,(z)}.$$

We can write $\theta_1\left(\frac{\pi z}{4K_3}, \frac{iR}{\sqrt{3}}\right) = \theta_1'\left(0, \frac{iR}{\sqrt{3}}\right) z + O(z^2)$ and $\theta_1\left(\frac{\pi z}{4K_3} + \frac{\pi}{3},\ \frac{iR}{\sqrt{3}}\right) = \theta_1\left(\frac{\pi}{3},\ \frac{iR}{\sqrt{3}}\right) + \frac{\pi}{4K_3}\theta_1'\left(\frac{\pi}{3},\ \frac{iR}{\sqrt{3}}\right) z + O\left(z^2\right)$ to get that

$$\mathrm{cm}(z + p_{m,n}) = \omega^{2n}\frac{4K_3}{\pi}\ \frac{\theta_1\left(\frac{\pi}{3},\ \frac{iR}{\sqrt{3}}\right)}{\theta_1'\left(0,\ \frac{iR}{\sqrt{3}}\right)}\ \frac{1}{z} + \omega^{2n}\frac{\theta_1'\left(\frac{\pi}{3},\ \frac{iR}{\sqrt{3}}\right)}{\theta_1'\left(0, \frac{iR}{\sqrt{3}}\right)} + O\left(z\right).$$

By definition of K_3, the $\frac{1}{z}$ coefficient reduces to ω^{2n}. The constant term is $\omega^{2n}\frac{\alpha}{2}$. Thus, we have that $\mathrm{cm}(z) = \frac{\omega^{2n}}{z-p_{m,n}} + \frac{\omega^{2n}\alpha}{2} + O(z - p_{m,n})$ so that $\mathrm{cm}^2(z) = \frac{\omega^n}{(z-p_{m,n})^2} + \frac{\omega^n\alpha}{z-p_{m,n}} + O\left(1\right)$. Using the fact $\mathrm{sm}(z) = -\frac{\omega^n}{z-p_{m,n}} + O\left(1\right)$, we have that $\mathrm{sm}'(z) = \frac{\omega^n}{(z-p_{m,n})^2} + O\left(1\right)$. Using these estimates, it can be shown that $H(z) = O(1)$ near all the poles of $\mathrm{sm}(z)$, so $H(z)$ must be constant. Evaluating at $z = 0$ and using the fact $\mathrm{sm}'(0) = 1$ gives H is identically 0, which proves the derivative formula for $\mathrm{sm}(z)$.

To find the derivative of $\mathrm{cm}(z)$, we use theorem 2.1 and implicitly differentiate the equation

$$\mathrm{sm}^3(z) + 3\alpha\mathrm{sm}(z)\mathrm{cm}(z) + \mathrm{cm}^3(z) = 1$$

and solve for $\mathrm{cm}'(z)$. Substituting the derivative of $\mathrm{sm}(z)$ and simplifying yields $\mathrm{cm}'(z) = -\mathrm{sm}^2(z) - \alpha\mathrm{cm}(z)$.

□

We see that the functions $y = \mathrm{sm}(z)$ and $x = \mathrm{cm}(z)$ are elliptic functions that parameterize the curve $x^3 + 3\alpha xy + y^3 = 1$ and satisfy the initial value problem $y' = x^2 + \alpha y$, $x' = -y^2 - \alpha x$, with $x(0) = 1$ and $y(0) = 0$. These are precisely the same functions studied by Dixon in [1] and are equivalent to his functions, the only difference being that the α parameter Dixon had considered is opposite in sign to ours. Furthermore, Dixon had provided addition formulae for these functions [1]:

$$\mathrm{sm}\left(u + v\right) = \frac{\mathrm{sm}^2\left(u\right)\mathrm{cm}\left(v\right) - \mathrm{cm}\left(u\right)\mathrm{sm}^2\left(v\right)}{\mathrm{sm}\left(u\right)\mathrm{cm}^2\left(v\right) - \mathrm{cm}^2\left(u\right)\mathrm{sm}\left(v\right)},$$

$$\mathrm{cm}\left(u + v\right) = \frac{\mathrm{sm}\left(u\right)\mathrm{cm}\left(u\right) - \mathrm{sm}\left(v\right)\mathrm{cm}\left(v\right)}{\mathrm{sm}\left(u\right)\mathrm{cm}^2\left(v\right) - \mathrm{cm}^2\left(u\right)\mathrm{sm}\left(v\right)}.$$

From the derivatives of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$, we can see that the inverse functions are given by the following equations:

$$z = \int_0^{\mathrm{sm}(z)} \frac{dt}{y^2\left(t, \alpha\right) + \alpha t} = \int_{\mathrm{cm}(z)}^1 \frac{dt}{y^2\left(t, \alpha\right) + \alpha t},$$

where $y = y(t, \alpha)$ satisfies $y^3 + 3\alpha ty + t^3 = 1$. This gives an integral formula for K_3 in terms of the parameter α:

$$K_3 = \frac{3}{4}\int_0^1 \frac{dt}{y^2\left(t, \alpha\right) + \alpha t}.$$

The inverse function had been known to Dixon, as he had found the derivatives of his elliptic functions. This formula allows us to shift from thinking about K_3 as a function of R to instead a function of α. In section X, we will find a much more elegant integral formula for K_3 as a function of α, which is closely related to the integral studied by Berndt, Garvan, and Bhargava relevant in Ramanujan's theory of elliptic functions of base 3.

3. The Degenerate Case

The curve $x^3 + 3\alpha xy + y^3 = 1$ reduces to the line $x + y = 1$ when $\alpha = 1$. Hence, the quarter period K_3 is given by

$$K_3 = \frac{3}{4}\int_0^1 \frac{dt}{y^2(t,1)+t} = \frac{3}{4}\int_0^1 \frac{dt}{(1-t)^2+t} = \frac{\pi}{2\sqrt{3}}.$$

Furthermore, the functions $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ reduce to ordinary trigonometric functions. For example, the inverse of $\mathrm{sm}(z)$ is given by

$$\int_0^z \frac{dt}{(1-t)^2+t} = \frac{2}{\sqrt{3}}\left(\arctan\left(\frac{2z-1}{\sqrt{3}}\right) + \frac{\pi}{6}\right),$$

from which it follows that $\mathrm{sm}(z) = \frac{1}{2} + \frac{\sqrt{3}}{2}\tan\left(\frac{\sqrt{3}z}{2} - \frac{\pi}{6}\right)$ and $\mathrm{cm}(z) = \frac{1}{2} - \frac{\sqrt{3}}{2}\tan\left(\frac{\sqrt{3}z}{2} - \frac{\pi}{6}\right)$ when $\alpha = 1$. Moreover, if $\alpha = 1$, then we see that $q = e^{-\frac{2\pi R}{\sqrt{3}}}$ is such that $q\prod_{n=1}^{\infty}\frac{(1-q^{9n})^3}{(1-q^n)^3} = 0$. However, the infinite product is nonzero for $\Re(R) > 0$. Thus, we require that $q = 0$, which occurs when $R = \infty$ (i.e., R is a complex number whose real part goes to $+\infty$). Thus, the degenerate case occurs when $R \to \infty$, where we have that $\lim_{R\to\infty}\alpha = 1$ and $\lim_{R\to\infty} K_3 = \frac{\pi}{2\sqrt{3}}$.

4. Pole Expansions, Weierstrass Function, and Fourier Series

Using the poles of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ along with their residues, we can expand the functions into a pole expansion giving the following theorem:

Theorem 4.1.

$$\mathrm{sm}(z) = \sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}\left(\frac{\omega^{2n}}{\frac{4K_3}{3}(3m+1) + \frac{4inK_3'}{\sqrt{3}}} - \frac{\omega^{n}}{z - \frac{4K_3}{3}(3m+2) - \frac{4inK_3'}{\sqrt{3}}}\right),$$

$$\mathrm{cm}(z) = \sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}\left(\frac{\omega^{2n}}{z - \frac{4K_3}{3}(3m+2) - \frac{4inK_3'}{\sqrt{3}}} + \frac{\omega^{2n}}{\frac{4K_3}{3}(3m+1) - \frac{4inK_3'}{\sqrt{3}}}\right).$$

We can also write the Weierstrass elliptic function via $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$, leading to the following theorem:

Theorem 4.2. *Let $\wp(z)$ denote the Weierstrass elliptic function with periods 1 and $\frac{iR}{\sqrt{3}}$, then $\wp(z) = 16K_3^2 \frac{\mathrm{cm}(4K_3 z)}{\mathrm{sm}^2(4K_3 z)} + 12\alpha^2 K_3^2$ with invariants $g_2 = \left(\frac{27\alpha^4}{4} - 6\alpha\right)(4K_3)^4$ and $g_3 = -\frac{1}{8}\left(27\alpha^6 - 36\alpha^3 + 8\right)(4K_3)^6$.*

Proof. It can be verified that $\wp(z) = 16K_3^2 \frac{\mathrm{cm}(4K_3 z)}{\mathrm{sm}^2(4K_3 z)} + 12\alpha^2 K_3^2$ satisfies the differential equation $\left(\frac{d\wp}{dz}\right)^2 = 4\wp^3 - \left(\frac{27\alpha^4}{4} - 6\alpha\right)(4K_3)^4 \wp - \left(\frac{9\alpha^3}{2} - \frac{27\alpha^6}{8} - 1\right)(4K_3)^6$, from which it follows that

$$g_2 = \left(\frac{27\alpha^4}{4} - 6\alpha\right)(4K_3)^4 = \frac{4\pi^4}{3}\left(1 + 240\sum_{n=1}^{\infty} \frac{n^3 e^{-\frac{2\pi n R}{\sqrt{3}}}}{1 - e^{-\frac{2\pi n R}{\sqrt{3}}}}\right),$$

$$g_3 = \left(\frac{9\alpha^3}{2} - \frac{27\alpha^6}{8} - 1\right)(4K_3)^6 = \frac{8\pi^6}{27}\left(1 - 504\sum_{n=1}^{\infty} \frac{n^5 e^{-\frac{2\pi n R}{\sqrt{3}}}}{1 - e^{-\frac{2\pi n R}{\sqrt{3}}}}\right).$$

□

P.L. Robinson previously identified this theorem in [4] and is deducible from Dixon's study of these functions in [1]. Once the derivatives of the Dixon elliptic functions have been established, the proof of the definition of the Weierstrass function in terms of the Dixon elliptic functions falls immediately.

Corollary 4.3. *The function $\frac{\mathrm{sm}(z)}{\mathrm{cm}^2(z)}$ has pole expansion*

$$\frac{\mathrm{sm}(z)}{\mathrm{cm}^2(z)} = -\frac{3}{4}\alpha^2 + \frac{1}{z^2} + \sum_{\substack{(m,n)\in\mathbb{Z}^2 \\ (m,n)\neq(0,0)}} \left[\frac{1}{\left(z + 4mK_3 + \frac{4inK_3'}{\sqrt{3}}\right)^2} - \frac{1}{\left(4mK_3 + \frac{4inK_3'}{\sqrt{3}}\right)^2}\right]$$

$$= -\frac{3}{4}\alpha^2 + \frac{\pi^2}{16K_3^2}\csc^2\left(\frac{\pi z}{4K_3}\right) - \frac{\pi^2}{48K_3^2}\left(1 - 24\sum_{n=1}^{\infty}\frac{n e^{-\frac{2\pi n R}{\sqrt{3}}}}{1 - e^{-\frac{2\pi n R}{\sqrt{3}}}}\right) + \frac{\pi^2}{4K_3^2}\sum_{n=1}^{\infty}\frac{1 - \cos\left(\frac{\pi z}{2K_3}\right)\cosh\left(\frac{2\pi n R}{\sqrt{3}}\right)}{\left(\cosh\left(\frac{2\pi n R}{\sqrt{3}}\right) - \cos\left(\frac{\pi z}{2K_3}\right)\right)^2}.$$

Proof. We use theorem 4.1 and the pole expansion of $\wp(z)$ to obtain the pole expansion of $\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)}$. The second equality is obtained by summing over the index m in the pole expansion using the pole expansion of $\csc^2(z)$. □

Corollary 4.4. *The function $\frac{1}{\mathrm{sm}(z)}$ has the hyperbolic function expansion*

$$\frac{1}{\mathrm{sm}(z)} = \frac{\pi}{4\sqrt{3}K_3} + \frac{\pi}{4K_3}\cot\left(\frac{\pi z}{4K_3}\right) + \frac{\pi}{2K_3}\sum_{n=1}^{\infty}\frac{\sin\left(\frac{\pi z}{2K_3} + \frac{2\pi n}{3}\right) - \sin\left(\frac{2\pi n}{3}\right)e^{-\frac{2\pi n R}{\sqrt{3}}}}{\cosh\left(\frac{2\pi n R}{\sqrt{3}}\right) - \cos\left(\frac{\pi z}{2K_3}\right)}.$$

Proof. We note that $\frac{1}{\mathrm{sm}(z)} = \mathrm{cm}\left(z - \frac{4K_3}{3}\right)$ and use theorem 4.1 to find its pole expansion. Then using the same technique as that used in corollary 4.3, where instead we use the pole expansion of $\cot(z)$, to rewrite the pole expansion as a single sum over hyperbolic functions. □

This procedure of reducing a pole expansion of a function into an expansion consisting of hyperbolic functions is essential to obtaining the Fourier series of the function of interest, as the summand becomes periodic and has a Fourier series. This is exactly what we will do next to find the Fourier series of $\mathrm{sm}(z)$, $\mathrm{cm}(z)$, and various combinations of the two functions.

Theorem 4.5. *Let $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. We have the following Fourier series:*

$$\mathrm{sm}(z) = \frac{\pi}{4\sqrt{3}K_3} - \frac{\pi}{4K_3}\cot\left(\frac{\pi z}{4K_3} + \frac{\pi}{3}\right) + \frac{\pi}{K_3}\sum_{n=1}^{\infty}\frac{q^n\left(q^n\sin\left(\frac{2\pi n}{3} + \frac{\pi n z}{2K_3}\right) - \sin\left(\frac{2\pi n}{3} + \frac{\pi n z}{2K_3} - \frac{2\pi}{3}\right)\right)}{1 + q^n + q^{2n}},$$

$$\mathrm{cm}(z) = \frac{\pi}{4\sqrt{3}K_3} + \frac{\pi}{4K_3}\cot\left(\frac{\pi z}{4K_3} + \frac{\pi}{3}\right) + \frac{\pi}{K_3}\sum_{n=1}^{\infty}\frac{q^n\left(\sin\left(\frac{2\pi n}{3} + \frac{\pi n z}{2K_3} + \frac{2\pi}{3}\right) - q^n\sin\left(\frac{2\pi n}{3} - \frac{\pi n z}{2K_3}\right)\right)}{1 + q^n + q^{2n}},$$

$$\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)} = -\frac{3}{4}\alpha^2 + \frac{\pi^2}{16K_3^2}\csc^2\left(\frac{\pi z}{4K_3}\right) - \frac{\pi^2}{48K_3^2}\left(1 - 24\sum_{n=1}^{\infty}\frac{ne^{-\frac{2\pi n R}{\sqrt{3}}}}{1 - e^{-\frac{2\pi n R}{\sqrt{3}}}}\right) - \frac{\pi^2}{2K_3^2}\sum_{n=1}^{\infty}\frac{ne^{-\frac{2\pi n R}{\sqrt{3}}}\cos\left(\frac{\pi n z}{2K_3}\right)}{1 - e^{-\frac{2\pi n R}{\sqrt{3}}}},$$

$$\frac{\mathrm{sm}^2(z)}{\mathrm{cm}(z)} = -\alpha + \frac{\sqrt{3}\pi}{8K_3}\csc\left(\frac{\pi}{3} + \frac{\pi z}{4K_3}\right)\csc\left(\frac{\pi}{3} - \frac{\pi z}{4K_3}\right) + \frac{\sqrt{3}\pi}{K_3}\sum_{n=1}^{\infty}\frac{\chi_3(n)\,q^n\cos\left(\frac{\pi n z}{2K_3}\right)}{1 - q^n},$$

where $\chi_3(n)$ is the unique nontrivial Dirichlet character modulo 3.

Proof. To prove the Fourier series of $\mathrm{cm}(z)$ and $\mathrm{sm}(z)$, it suffices to find the Fourier series of $\frac{1}{\mathrm{sm}(z)}$ since $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ are just shifts and reflections of $\frac{1}{\mathrm{sm}(z)}$. We start by considering the hyperbolic function expansion of $\frac{1}{\mathrm{sm}(z)}$

$$\frac{1}{\mathrm{sm}(z)} = \frac{\pi}{4\sqrt{3}K_3} + \frac{\pi}{4K_3}\cot\left(\frac{\pi z}{4K_3}\right) + \frac{\pi}{2K_3}\sum_{n=1}^{\infty}\frac{\sin\left(\frac{\pi z}{2K_3} + \frac{2\pi n}{3}\right) - \sin\left(\frac{2\pi n}{3}\right)e^{-\frac{2\pi n R}{\sqrt{3}}}}{\cosh\left(\frac{2\pi n R}{\sqrt{3}}\right) - \cos\left(\frac{\pi z}{2K_3}\right)}.$$

The summand is rational function of $\cos\left(\frac{\pi z}{2K_3}\right)$ and $\sin\left(\frac{\pi z}{2K_3}\right)$. Thus, it is a rational function of $e^{\frac{i\pi z}{2K_3}}$. We can find the Fourier series of the summand by finding the power series expansion of the rational function, giving

$$\frac{\sin\left(\frac{\pi z}{2K_3} + \frac{2\pi n}{3}\right) - \sin\left(\frac{2\pi n}{3}\right)e^{-\frac{2\pi n R}{\sqrt{3}}}}{\cosh\left(\frac{2\pi n R}{\sqrt{3}}\right) - \cos\left(\frac{\pi z}{2K_3}\right)} = 2\sum_{m=1}^{\infty}e^{-\frac{2\pi m n R}{\sqrt{3}}}\sin\left(\frac{2\pi n}{3} + \frac{\pi m z}{2K_3}\right).$$

We then sum over n and switch the order of summation to obtain, so that we sum over n first. This gives us the Fourier series of $\frac{1}{\mathrm{sm}(z)}$, from which the Fourier series of $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$ follow.

To find the Fourier series of $\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)}$, we consider the expansion

$$\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)} = -\frac{3}{4}\alpha^2 + \frac{\pi^2}{16K_3^2}\csc^2\left(\frac{\pi z}{4K_3}\right) - \frac{\pi^2}{48K_3^2}\left(1 - 24\sum_{n=1}^{\infty}\frac{ne^{-\frac{2\pi n R}{\sqrt{3}}}}{1 - e^{-\frac{2\pi n R}{\sqrt{3}}}}\right)$$

$$+\frac{\pi^2}{4K_3^2}\sum_{n=1}^{\infty}\frac{1-\cos\left(\frac{\pi z}{2K_3}\right)\cosh\left(\frac{2\pi nR}{\sqrt{3}}\right)}{\left(\cosh\left(\frac{2\pi nR}{\sqrt{3}}\right)-\cos\left(\frac{\pi z}{2K_3}\right)\right)^2}$$

We apply the same method as before and find the Fourier series expansion of the summand in the last term:

$$\frac{1-\cos\left(\frac{\pi z}{2K_3}\right)\cosh\left(\frac{2\pi nR}{\sqrt{3}}\right)}{\left(\cosh\left(\frac{2\pi nR}{\sqrt{3}}\right)-\cos\left(\frac{\pi z}{2K_3}\right)\right)^2}=-2\sum_{m=1}^{\infty}me^{-\frac{2\pi mnR}{\sqrt{3}}}\cos\left(\frac{\pi mz}{2K_3}\right).$$

Summing over n and switching the order of summation gives the Fourier series for $\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)}$. The final Fourier series follows by taking the logarithmic derivative of the infinite product expression for $\mathrm{cm}(z)$. □

Evaluating these Fourier series or their derivatives/integrals at certain values of z yield inversion formulae for various Lambert-type series when $q=e^{-\frac{2\pi R}{\sqrt{3}}}=e^{-\frac{2\pi}{\sqrt{3}}\frac{K_3'}{K_3}}$. That is, they give formulas for such series in terms of α and K_3. These formulas are the"α-K formulas" for the theory of Dixon elliptic functions, which are analogous to K-formulas in the theory of Jacobi's elliptic functions.

Corollary 4.6. *Let $q=e^{-\frac{2\pi R}{\sqrt{3}}}=e^{-\frac{2\pi}{\sqrt{3}}\frac{K_3'}{K_3}}$ with $\Re(R)>0$. Then,*

$$\frac{2\sqrt{3}K_3}{\pi}=1+6\sum_{n=1}^{\infty}\frac{q^n\cos\left(\frac{4\pi n}{3}\right)}{1+q^n+q^{2n}},$$

$$\frac{2\sqrt{3}\alpha K_3}{\pi}=1+6\sum_{n=0}^{\infty}\left(\frac{q^{3n+1}}{1-q^{3n+1}}-\frac{q^{3n+2}}{1-q^{3n+2}}\right),$$

$$\frac{24\sqrt{3}K_3^3}{\pi^3}=1+6\sum_{n=1}^{\infty}\left(\frac{(3n-1)^2q^{3n-1}}{1-q^{3n-1}}-\frac{(3n-2)^2q^{3n-2}}{1-q^{3n-2}}\right),$$

$$\frac{12\alpha^2K_3^2}{\pi^2}=1+12\sum_{n=0}^{\infty}\left(\frac{(3n+1)q^{3n+1}}{1-q^{3n+1}}+\frac{(3n+2)q^{3n+2}}{1-q^{3n+2}}\right),$$

$$\frac{144\alpha^4K_3^4}{\pi^4}=1+24\sum_{n=1}^{\infty}n^3\left(\frac{q^n}{1-q^n}+\frac{9q^{3n}}{1-q^{3n}}\right).$$

Proof. The first α-K-formula comes evaluating the Fourier series of $\mathrm{sm}(z)+\mathrm{cm}(z)$, which can easily be obtained via theorem 4.5, at $z=\frac{4K_3}{3}$. The second formula can be found by evaluating the Fourier series for $\frac{\mathrm{sm}^2(z)}{\mathrm{cm}(z)}$ at $z=0$. For the third formula, we consider the Fourier series of $\frac{d^2}{dz^2}\frac{\mathrm{sm}^2(z)}{\mathrm{cm}(z)}$ and evaluate at $z=0$. The third formula is obtained by evaluating the Fourier

series of $\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)}$ at $z = \frac{4K_3}{3}$. The proof of the last formula is less direct than the other 4. We evaluate the Fourier series for $\frac{d^2}{dz^2}\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)}$ at $z = \frac{4K_3}{3}$ to get

$$\frac{144\alpha K_3^4}{\pi^4} = 1 + 3\sum_{n=1}^{\infty}\left(\frac{81n^3q^{3n}}{1-q^{3n}} - \frac{n^3q^n}{1-q^n}\right).$$

Then, we can use theorem 4.2 to get

$$\frac{144\alpha K_3^4}{\pi^4} = \frac{162\alpha^4 K_3^4}{\pi^4} - \frac{1}{8}\left(1 + 240\sum_{n=1}^{\infty}\frac{n^3q^n}{1-q^n}\right),$$

which gives the desired result. □

Corollary 4.6 implies various series relationships such as

$$\left[1 + 6\sum_{n=0}^{\infty}\left(\frac{x^{3n+1}}{1-x^{3n+1}} - \frac{x^{3n+2}}{1-x^{3n+2}}\right)\right]^2 = 1+12\sum_{n=0}^{\infty}\left(\frac{(3n+1)\,x^{3n+1}}{1-x^{3n+1}} + \frac{(3n+2)\,x^{3n+2}}{1-x^{3n+2}}\right),$$

$$\left[1 + 6\sum_{n=0}^{\infty}\left(\frac{x^{3n+1}}{1-x^{3n+1}} - \frac{x^{3n+2}}{1-x^{3n+2}}\right)\right]^4 = 1 + 24\sum_{n=1}^{\infty}\left(\frac{n^3x^n}{1-x^n} + \frac{9n^3x^{3n}}{1-x^{3n}}\right),$$

$$\left[1 + 6\sum_{n=1}^{\infty}\frac{x^n\cos\left(\frac{4\pi n}{3}\right)}{1+x^n+x^{2n}}\right]^3 = 1+6\sum_{n=1}^{\infty}\left(\frac{(3n-1)^2\,x^{3n-1}}{1-x^{3n-1}} - \frac{(3n-2)^2\,x^{3n-2}}{1-x^{3n-2}}\right).$$

All of these series relations had been known via Ramanujan's theory of base 3 elliptic functions. However, we were able to prove the same using the Fourier series of Dixon elliptic functions. The first of these relations had been given by Berndt, Garvan, and Bhargava in [5]. The other two relations were given by S. Cooper in [3].

The importance of these series relations is given by the fact

$$\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} x^{m^2+mn+n^2} = 1 + 6\sum_{n=0}^{\infty}\left(\frac{x^{3n+1}}{1-x^{3n+1}} - \frac{x^{3n+2}}{1-x^{3n+2}}\right).$$

This identity was first found by Ramanujan and a proof is provided by Hirschhorn [6]. Thus, we see that these series relations resulting from corollary 4.6 reveal formulae for the number of ways a number can be represented as sums of numbers of the form $m^2 + mn + n^2$ in terms of certain divisor functions.

5. Modular Equations and The Borweins' Cubic Theta Functions

The goal of this section is to re-prove many of the major results in Ramanujan's theory of elliptic function of base 3 given by Berndt, Bhargava, and Garvan in [5] using the generalized Dixon elliptic functions. As such, most of the results in this section was already established by Berndt, Bhargava, and Garvan, but will be proved independent of their approach.

Definition 5.1. Let $a(q) = 1+6\sum_{n=0}^{\infty}\left(\frac{q^{3n+1}}{1-q^{3n+1}} - \frac{q^{3n+2}}{1-q^{3n+2}}\right) = 1+6\sum_{n=1}^{\infty}\frac{q^n}{1+q^n+q^{2n}}$, $b(q) = 1+6\sum_{n=1}^{\infty}\frac{q^n\cos\left(\frac{4\pi n}{3}\right)}{1+q^n+q^{2n}}$, and $c(q) = 2\sum_{n=1}^{\infty}\frac{q^{\frac{n}{3}}\left(1-\cos\left(\frac{4\pi n}{3}\right)\right)}{1+q^{\frac{n}{3}}+q^{\frac{2n}{3}}}$.

Theorem 5.2 (Modular Equations of Degree 3)**.** *Let* $\alpha = \alpha(R) = 1 + 9q\prod_{n=1}^{\infty}\frac{\left(1-q^{9n}\right)^3}{\left(1-q^n\right)^3}$ *and* $K_3 = K_3(R) = \frac{\pi}{2\sqrt{3}}\prod_{n=1}^{\infty}\frac{(1-q^n)^3}{1-q^{3n}}$, *where* $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. *Furthermore, let* $\beta = \alpha(3R)$ *and* $L_3 = K_3(3R)$. *Then,*

$$\beta = \frac{\alpha+2}{\sqrt[3]{9\left(\alpha^2+\alpha+1\right)}},$$

$$L_3 = \sqrt[3]{\frac{\alpha^2+\alpha+1}{3}}K_3.$$

Proof. We start by noticing that $b(q) = \frac{1}{2}\left(3a\left(q^3\right) - a\left(q\right)\right)$, which can be found by splitting the index of the series for $b(q)$ modulo 3. Likewise, $c(q) = \frac{1}{2}\left(a\left(q^{\frac{1}{3}}\right) - a\left(q\right)\right)$. For $q = e^{-\frac{2\pi R}{\sqrt{3}}}$, we have $b(q) = \frac{2\sqrt{3}K_3}{\pi}$, $a(q) = \frac{2\sqrt{3}\alpha K_3}{\pi}$, and $a(q^3) = \frac{2\sqrt{3}\beta L_3}{\pi}$ using corollary 4.6. Thus, one modular equation of degree 3 is

$$\beta L_3 = \frac{\alpha+2}{3}K_3.$$

For the second modular equation of degree 3, we consider the Fourier series

$$G\left(z\right) = -\frac{1}{6} + \frac{K_3}{\sqrt{3}\pi}\left(\frac{1}{\operatorname{sm}\left(4K_3z\right)} + \frac{1}{\operatorname{sm}\left(-4K_3z\right)}\right) = \sum_{n=1}^{\infty}\frac{q^n\cos\left(2\pi nz\right)}{1+q^n+q^{2n}},$$

which can be proved using the Fourier series of $\operatorname{sm}(z)$ from theorem 4.5. Next, we consider the Fourier series for $\frac{1}{3}\left(G\left(\frac{z}{3}\right) + G\left(\frac{z+1}{3}\right) + G\left(\frac{z+2}{3}\right)\right)$, which picks out the terms of the Fourier series above where $n \equiv 0\,(\mathrm{mod}3)$:

$$\frac{1}{3}\left(G\left(\frac{z}{3}\right) + G\left(\frac{z+1}{3}\right) + G\left(\frac{z+2}{3}\right)\right) = \sum_{n=1}^{\infty}\frac{q^{3n}\cos\left(2\pi nz\right)}{1+q^{3n}+q^{6n}}.$$

Evaluating at $z = \frac{2}{3}$ gives

$$1+6\sum_{n=1}^{\infty}\frac{q^{3n}\cos\left(\frac{4\pi n}{3}\right)}{1+q^{3n}+q^{6n}} = \frac{2K_3}{\sqrt{3}\pi}\left(\operatorname{sm}\left(\frac{8K_3}{9}\right) + \operatorname{sm}\left(\frac{20K_3}{9}\right) + \operatorname{sm}\left(\frac{32K_3}{9}\right)\right).$$

On one hand, by definition of L_3, we have that $1+6\sum_{n=1}^{\infty}\frac{q^{3n}\cos\left(\frac{4\pi n}{3}\right)}{1+q^{3n}+q^{6n}} = \frac{2\sqrt{3}L_3}{\pi}$. Computing the sum of the values sm is quite harder, but it can be done. We start by noticing that $\operatorname{sm}\left(\frac{20K_3}{9}\right) = \frac{1}{\operatorname{cm}\left(\frac{8K_3}{9}\right)}$ and $sm\left(\frac{32K_3}{9}\right) = -\frac{cm\left(\frac{8K_3}{9}\right)}{\operatorname{sm}\left(\frac{8K_3}{9}\right)}$. Let $r_1 = \operatorname{sm}\left(\frac{8K_3}{9}\right)$, $r_2 = \frac{1}{\operatorname{cm}\left(\frac{8K_3}{9}\right)}$, and $r_3 = -\frac{\operatorname{cm}\left(\frac{8K_3}{9}\right)}{\operatorname{sm}\left(\frac{8K_3}{9}\right)}$. We can see that $r_1r_2r_3 = -1$ so that $r_3 = -\frac{1}{r_1r_2}$. Using addition formulas for $\operatorname{sm}(z)$ and $\operatorname{cm}(z)$, it may be seen that

$$\mathrm{sm}(3u) = \mathrm{sm}\,(u)\,\mathrm{cm}\,(u)\,\frac{1+\mathrm{sm}^3\,(u)+\mathrm{cm}^3\,(u)+\mathrm{sm}^6\,(u)+\mathrm{cm}^6\,(u)-\mathrm{sm}^3\,(u)\,\mathrm{cm}^3\,(u)}{\mathrm{cm}^3\,(u)+3\mathrm{cm}^3\,(u)\,\mathrm{sm}^3\,(u)+\mathrm{cm}^6\,(u)\,\mathrm{sm}^3\,(u)-\mathrm{sm}^6\,(u)}.$$

When $u = \frac{8K_3}{9},\ \frac{20K_3}{9},\ \frac{32K_3}{9}$, $\mathrm{sm}(3u)$ is not defined (i.e., it has a pole there). Since the numerator of the triple angle formula is not in general 0, we have that

$$\mathrm{cm}^3\left(\frac{8K_3}{9}\right)+3\mathrm{cm}^3\left(\frac{8K_3}{9}\right)\mathrm{sm}^3\left(\frac{8K_3}{9}\right)+\mathrm{cm}^6\left(\frac{8K_3}{9}\right)\mathrm{sm}^3\left(\frac{8K_3}{9}\right)-\mathrm{sm}^6\left(\frac{8K_3}{9}\right)=0$$

Dividing the equation by $\mathrm{cm}^6\left(\frac{8K_3}{9}\right)$, we get

$$-r_1^6r_2^6+3r_1^3r_2^3+r_2^3+r_1^3=0.$$

Let $P = r_1r_2$ and $S = r_1 + r_2$. Then, the equation immediately above is equivalent to $\left(S-P^2\right)\left(S^2+SP^2+P^4-3P\right)=0$. It may be seen that the second factor is not in general 0 for all R, so this forces the first factor to be 0. Thus, $S = P^2$ and we have $r_1+r_2=(r_1r_2)^2$ and $r_3=-\frac{1}{r_1r_2}$. We then get that

$$r_1r_2+r_2r_3+r_3r_1 = r_1r_2-\frac{r_1+r_2}{r_1r_2} = P-\frac{S}{P} = \frac{P^2-S}{P} = 0.$$

Since $r_1r_2 = P$ and $r_1 + r_2 = P^2$, we have that r_1 and r_2 satisfy the quadratic $t^2-P^2t+P=0$. In particular, $r_2^3=P^2r_2^2-Pr_2$ and $r_2^2=P^2r_2-P$. We also have that $\mathrm{sm}^3\left(\frac{4K_3}{9}\right)+3\alpha\mathrm{sm}\left(\frac{4K_3}{9}\right)\mathrm{cm}\left(\frac{4K_3}{9}\right)+\mathrm{cm}^3\left(\frac{4K_3}{9}\right)=1$, which is equivalent to $r_1^3r_2^3+3\alpha r_1r_2^2+1=P^3+3\alpha Pr_2+1=r_2^3$. Thus, $r_2^3=P^2r_2^2-Pr_2=P^2\left(P^2r_2-P\right)-Pr_2=\left(P^4-P\right)r_2-P^3$, which gives $P^3+3\alpha Pr_2+1=\left(P^4-P\right)r_2-P^3$. We can solve for r_2 to get

$$r_2=-\frac{2P^3+1}{P\left(3\alpha+1-P^3\right)}.$$

We substitute this expression for r_2 into the quadratic equation $r_2^2-P^2r_2+P=0$. Upon simplification, we arrive at the equation

$$P^9-3P^6-\left(9\alpha^2+9\alpha+6\right)P^3-1=0,$$

which is a cubic in P^3. This equation is the key equation that allows us to solve for $r_1+r_2+r_3$. Next, we have $r_1+r_2+r_3=P^2-\frac{1}{r_1r_2}=P^2-\frac{1}{P}$. Cubing both sides of this equation, we get $\left(r_1+r_2+r_3\right)^3=P^6-3P^3+3-\frac{1}{P^3}=\frac{P^9-3P^6+3P^3-1}{P^3}$. The cubic satisfied by P^3 implies that $P^9-3P^6+3P^3-1=9\left(\alpha^2+\alpha+1\right)P^3$. Thus, $r_1+r_2+r_3=\sqrt[3]{9\left(\alpha^2+\alpha+1\right)}$, which is the value we wanted to find. This gives our next modular equation, which is

$$\frac{2\sqrt{3}L_3}{\pi}=\frac{2\sqrt{3}K_3}{\pi}\sqrt[3]{\frac{\alpha^2+\alpha+1}{3}}.$$

Thus, our two modular equations are

$$\beta L_3 = \frac{\alpha+2}{3} K_3,$$

$$L_3 = \sqrt[3]{\frac{\alpha^2+\alpha+1}{3}} K_3.$$

Solving the system for β and L_3 in terms of α and K_3 gives the desired result. □

Theorem 5.3 (Modular Equations of Degree 2)**.** *Let β and L_3 be degree 2 over α and K_3 respectively, so that $\beta = \alpha(2R)$ and $L_3 = K_3(2R)$. Let $g(\alpha) = \sqrt[3]{1+\sqrt{1-\alpha^3}} + \sqrt[3]{1-\sqrt{1-\alpha^3}} = 2\sqrt{\alpha}\cos\left(\frac{\arctan\sqrt{\alpha^3-1}}{3}\right)$. Then,*

$$\beta = \frac{1}{g(\alpha)}\left(\frac{2}{g(\alpha)} + \alpha\right),$$

$$L_3 = \frac{g(\alpha)}{2} K_3.$$

Proof. From the product definition of $\mathrm{cm}(z)$, we can see that $\mathrm{cm}(2K_3) = -1$. Thus, $x = \mathrm{sm}(2K_3) = \frac{1}{\mathrm{sm}(2K_3/3)}$ satisfies $x^3 - 3\alpha x - 2 = 0$. Thus, $\mathrm{sm}(2K_3) = \sqrt[3]{1+\sqrt{1-\alpha^3}} + \sqrt[3]{1-\sqrt{1-\alpha^3}} = 2\sqrt{\alpha}\cos\left(\frac{\arctan\left(\sqrt{\alpha^3-1}\right)}{3}\right)$.

Let $\beta = \alpha(2R)$, $L_3 = K_3(2R)$, and $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. We use the Fourier series of $\mathrm{sm}(z)$ from theorem 4.5 to get

$$\frac{4\sqrt{3}K_3}{\pi \mathrm{sm}(2K_3)} = 1 + 6\sum_{n=1}^{\infty} \frac{(-1)^n q^n}{1+q^n+q^{2n}}$$

$$= 2\left(1 + 6\sum_{n=1}^{\infty} \frac{q^{2n}}{1+q^{2n}+q^{4n}}\right) - \left(1 + 6\sum_{n=1}^{\infty} \frac{q^n}{1+q^n+q^{2n}}\right).$$

Therefore,

$$\frac{2\sqrt{3}\beta L_3}{\pi} = 1 + 6\sum_{n=1}^{\infty} \frac{q^{2n}}{1+q^{2n}+q^{4n}} = \frac{4\sqrt{3}K_3}{\pi \mathrm{sm}(2K_3)} + \frac{3\alpha K_3}{\pi}.$$

Thus, our first modular equation is

$$\beta L_3 = \left(\frac{1}{\mathrm{sm}(2K_3)} + \frac{\alpha}{2}\right) K_3.$$

Next, we use the Fourier series of $\frac{1}{\mathrm{sm}(z)}$, which can be found using theorem 4.5, to get

$$\frac{\sqrt{3}K_3}{\pi}\left(\frac{1}{\mathrm{sm}(2K_3 z)} + \frac{1}{\mathrm{sm}(2K_3(z+1))} + \frac{1}{\mathrm{sm}(-2K_3 z)} + \frac{1}{\mathrm{sm}(-2K_3(z+1))}\right) = 1 + 6\sum_{n=1}^{\infty} \frac{q^{2n}\cos(2\pi n z)}{1+q^{2n}+q^{4n}}.$$

Evaluating at $z = \frac{2K_3}{3}$ gives

$$\frac{\sqrt{3}K_3}{\pi}\mathrm{sm}\left(2K_3\right) = 1 + 6\sum_{n=1}^{\infty}\frac{q^{2n}\cos\left(\frac{4\pi n}{3}\right)}{1+q^{2n}+q^{4n}} = \frac{2\sqrt{3}L_3}{\pi}.$$

This gives us the following system of equations:

$$\beta L_3 = \left(\frac{1}{\mathrm{sm}\left(2K_3\right)} + \frac{\alpha}{2}\right) K_3,$$

$$L_3 = \frac{\mathrm{sm}\left(2K_3\right)}{2}K_3.$$

Solving the system and letting $g(\alpha) = \mathrm{sm}(2K_3)$ proves the theorem. □

An equivalent version of the degree 3 modular equations in terms of ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$ was known to Ramanujan. A proof of this modular equation under the formulation of ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$ can be found in [5].

Lemma 5.4. *Let* $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. *Then,* $c(q) = \frac{2\sqrt{3}\left(\alpha^3-1\right)^{\frac{1}{3}}K_3}{\pi}$.

Proof. From the definition of $c(q)$, we have that $c(q) = \frac{1}{2}\left(a\left(q^{\frac{1}{3}}\right) - a\left(q\right)\right)$. Thus, we let $q = e^{-\frac{2\pi R}{\sqrt{3}}}$ and use theorem 5.2 to get

$$c(q^3) = \frac{2}{\sqrt{3}\pi}\left(\alpha - 1\right)K_3.$$

Since $\beta = \frac{\alpha+2}{\sqrt[3]{9(\alpha^2+\alpha+1)}}$ and $L_3 = \sqrt[3]{\frac{\alpha^2+\alpha+1}{3}}K_3$, we have that $\left(\alpha-1\right)K_3 = 3\sqrt[3]{\beta^3-1}L_3$ and $c(q^3) = \frac{2\sqrt{3}}{\pi}\left(\beta^3-1\right)^{\frac{1}{3}}L_3$. Thus, $c(q) = \frac{2\sqrt{3}}{\pi}\left(\alpha^3-1\right)^{\frac{1}{3}}K_3$. □

Theorem 5.5 (J.M. Borwein, P.B. Borwein). *Let* $|q| < 1$. *Then,*

$$b^3(q) + c^3(q) = a^3(q).$$

Proof. We let $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. Then, by corollary 4.6 and lemma 5.4, we have

$$b^3(q)+c^3(q) = \left(\frac{2\sqrt{3}K_3}{\pi}\right)^3 + \left(\frac{2\sqrt{3}\left(\alpha^3-1\right)^{\frac{1}{3}}K_3}{\pi}\right)^3 = \left(\frac{2\sqrt{3}\alpha K_3}{\pi}\right)^3 = a^3\left(q\right).$$

Since $\Re(R) > 0$, we have that $|q| < 1$. □

The functions $a(q)$, $b(q)$, and $c(q)$ are the Borwein cubic theta functions and this elegant identity had been established by J.M. and P.B. Borwein in [7]. The cubic theta functions play the role of the Jacobi theta functions $\theta_2(0,\tau)$, $\theta_2(0,\tau)$, $\theta_4(0,\tau)$, which satisfy $\theta_2^4\left(0,\tau\right) + \theta_4^4\left(0,\tau\right) = \theta_3^4\left(0,\tau\right)$. The Borwein brothers initially defined their functions $a(q)$, $b(q)$, and $c(q)$ as

$$a\left(q\right) = \sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} q^{m^2+mn+n^2},$$

$$b\left(q\right)=\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}e^{\frac{2\pi i(m-n)}{3}}q^{m^2+mn+n^2},$$

$$c\left(q\right)=\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}q^{\left(m+\frac{1}{3}\right)^2+\left(m+\frac{1}{3}\right)\left(n+\frac{1}{3}\right)+\left(n+\frac{1}{3}\right)^2}.$$

Ramanujan had already given the equivalence of our definition of $a(q)$ and the Borweins' definition of $a(q)$. Furthermore, the Borwein brothers also showed [7] that

$$\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}e^{\frac{2\pi i(m-n)}{3}}q^{m^2+mn+n^2}=\frac{1}{2}\left(3a\left(q^3\right)-a\left(q\right)\right),$$

$$\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}q^{\left(m+\frac{1}{3}\right)^2+\left(m+\frac{1}{3}\right)\left(n+\frac{1}{3}\right)+\left(n+\frac{1}{3}\right)^2}=\frac{1}{2}\left(a\left(q^{\frac{1}{3}}\right)-a\left(q\right)\right),$$

which shows that our definition of $b(q)$ and $c(q)$ is equivalent to the Borweins' definitions.

Corollary 5.6. *Let $|q|<1$. Then,*

$$\left(\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}q^{(3m+1)^2+(3m+1)(3n+1)+(3n+1)^2}\right)^2=\sum_{n=0}^{\infty}3\sigma\left(3n+2\right)q^{3(3n+2)},$$

where $\sigma(n)=\sum_{d|n}d$ is the sum of divisors function.

Proof. □

Corollary 5.6 was also given by S. Cooper in [3]. This result implies a formula for the numbers of ways a natural number can be written as two numbers, where each of these 2 numbers is of the form $(3m+1)^2+(3n+1)(3m+1)+(3n+1)^2$ in terms of $\sigma(n)$.

6. Reparameterization into the Jacobi Framework

In this section, we will essentially find a way to describe the Dixon elliptic functions and K_3 as functions of α. We will see that α will play the same role as the modulus k for the Jacobi theta functions. We will also see that, just as we have the complementary modulus k', we will also have α'. We will then give a simple expressions for K_3 and K_3' as functions of α in terms of Ramanujan's function ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$.

Theorem 6.1. *Let $\Re(R)>0$, $q=e^{-\frac{2\pi R}{\sqrt{3}}}$, $\alpha(R)=1+9q\prod_{n=1}^{\infty}\frac{\left(1-q^{9n}\right)^3}{\left(1-q^n\right)^3}$, and $K_3(R)=\frac{\pi}{2\sqrt{3}}\prod_{n=1}^{\infty}\frac{(1-q^n)^3}{1-q^{3n}}$. Then,*

$$\alpha\left(\frac{1}{R}\right)=\frac{\alpha\left(R\right)}{\sqrt[3]{\alpha^3\left(R\right)-1}},$$

$$K_3\left(\frac{1}{R}\right)=\sqrt[3]{\alpha^3\left(R\right)-1}RK_3\left(R\right)=\sqrt[3]{\alpha^3\left(R\right)-1}K_3'\left(R\right).$$

Proof. Using corollary 4.6, we have

$$\frac{144\alpha^4 K_3^4}{\pi^4} = a^4\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right) = \frac{E_4\left(\frac{iR}{\sqrt{3}}\right) + 9E_4\left(i\sqrt{3}R\right)}{10},$$

where $E_4(\tau) = 1 + 240\sum_{n=1}^{\infty}\frac{n^3 e^{2\pi i n\tau}}{1-e^{2\pi i n\tau}}$ is the weight 4 Eisenstein series. Let $\tau = \frac{iR}{\sqrt{3}}$. Then, we see that $R \to \frac{1}{R}$ corresponds to $\tau \to -\frac{1}{3\tau}$. Furthermore,

$$\frac{E_4\left(-\frac{1}{3\tau}\right) + 9E_4\left(3\left(-\frac{1}{3\tau}\right)\right)}{10} = \frac{81\tau^4 E_4(3\tau) + 9\tau^4 E_4(\tau)}{10} = 9\tau^4\frac{E_4(\tau) + 9E_4(3\tau)}{10}.$$

Thus, we have that

$$\frac{144\alpha\left(\frac{1}{R}\right)K_3\left(\frac{1}{R}\right)}{\pi^4} = R^4\frac{144\alpha(R)K_3(R)}{\pi^4}.$$

That is, $a\left(e^{-\frac{2\pi}{\sqrt{3}R}}\right) = Ra\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)$, which is equivalent to $\frac{\alpha\left(\frac{1}{R}\right)K_3\left(\frac{1}{R}\right)}{\alpha(R)K_3(R)} = R$. Once again, we use corollary 4.6 to get that

$$b\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right) = \frac{1}{2}\left(3a\left(e^{-2\sqrt{3}\pi R}\right) - a\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)\right) = \frac{2\sqrt{3}K_3(R)}{\pi}.$$

Thus, $b\left(e^{-\frac{2\pi}{\sqrt{3}R}}\right) = \frac{1}{2}\left(3a\left(e^{-\frac{2\sqrt{3}\pi}{R}}\right) - a\left(e^{-\frac{2\pi}{\sqrt{3}R}}\right)\right)$. However, $a\left(e^{-\frac{2\pi}{\sqrt{3}R}}\right) = Ra\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)$ and $a\left(e^{-\frac{2\sqrt{3}\pi}{R}}\right) = \frac{R}{3}a\left(e^{-\frac{2\pi R}{3\sqrt{3}}}\right)$, so

$$b\left(e^{-\frac{2\pi}{\sqrt{3}R}}\right) = \frac{R}{2}\left(a\left(e^{-\frac{2\pi R}{3\sqrt{3}}}\right) - a\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)\right) = Rc\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right).$$

Using lemma 5.4, we get $b\left(e^{-\frac{2\pi}{\sqrt{3}R}}\right) = \frac{2\sqrt{3}R\left(\alpha^3(R)-1\right)^{\frac{1}{3}}K_3(R)}{\pi}$, which is equivalent to $K_3\left(\frac{1}{R}\right) = R\left(\alpha^3(R)-1\right)^{\frac{1}{3}}K_3(R)$. Thus, we have

$$\frac{\alpha\left(\frac{1}{R}\right)K_3\left(\frac{1}{R}\right)}{\alpha(R)K_3(R)} = R,$$

$$K_3\left(\frac{1}{R}\right) = R\left(\alpha^3(R)-1\right)^{\frac{1}{3}}K_3(R).$$

We solve the system to get $\alpha\left(\frac{1}{R}\right) = \frac{\alpha(R)}{\sqrt[3]{\alpha^3(R)-1}}$ and $K_3\left(\frac{1}{R}\right) = R\left(\alpha^3(R)-1\right)^{\frac{1}{3}}K_3(R)$, as desired. □

Theorem 6.1 allows us to better understand K_3 as a function of α and place this theory of the Dixon elliptic functions in a way analogous to Jacobi's elliptic functions. For the rest of this paper, unless stated otherwise, we write K_3 to mean K_3 as a function of α. We see that $R \to \frac{1}{R}$ corresponds to $\alpha \to \frac{\alpha}{\sqrt[3]{\alpha^3-1}}$. Thus, for the remainder of this paper, we define $\alpha' := \frac{\alpha}{\sqrt[3]{\alpha^3-1}}$, which is the "complementary elliptic modulus" (whereas α

is the modulus of the Dixon elliptic functions). We also then have that $K_3' = \frac{1}{\sqrt[3]{\alpha^3-1}} K_3(\alpha') = \frac{1}{\sqrt[3]{\alpha^3-1}} K_3\left(\frac{\alpha}{\sqrt[3]{\alpha^3-1}}\right)$ and $\frac{1}{\sqrt[3]{\alpha^3-1}} \frac{K_3\left(\frac{\alpha}{\sqrt[3]{\alpha^3-1}}\right)}{K_3(\alpha)} = R$.

We note that when $\alpha = 0$, we have that $K_3 = \frac{\sqrt{3}\Gamma^3\left(\frac{1}{3}\right)}{8\pi}$ and $q = -e^{-\frac{\pi}{\sqrt{3}}}$. Thus, we can use the modular equation of degree 2 from theorem 5.3 to determine that when $q = e^{-\frac{2\pi}{\sqrt{3}}}$, $K_3 = \frac{\sqrt{3}\sqrt[3]{2}\Gamma^3\left(\frac{1}{3}\right)}{16\pi}$. However, $q = e^{-\frac{2\pi}{\sqrt{3}}}$ corresponds to $R = 1$ and when $R = 1$, $\alpha = \alpha'$. Thus, we see that when $\alpha = \sqrt[3]{2}$, then $q = e^{-\frac{2\pi}{\sqrt{3}}}$, $R = 1$, and $K_3 = K_3' = \frac{\sqrt{3}\sqrt[3]{2}\Gamma^3\left(\frac{1}{3}\right)}{16\pi}$.

Theorem 6.2. *Let $\alpha' = \frac{\alpha}{\sqrt[3]{\alpha^3-1}}$, $K_3 = K_3(\alpha)$ be a function of α, and $q = e^{-\frac{2\pi R}{\sqrt{3}}} = e^{-\frac{2\pi}{\sqrt{3}}\frac{K_3'}{K_3}}$. Then,*

$$\frac{d\alpha'}{d\alpha} = -\frac{\alpha'^4}{\alpha^4},$$

$$\frac{dK_3}{d\alpha} = -\frac{\alpha'^3 K_3}{4\alpha} + \frac{\pi^2\alpha'^3}{48\alpha^3 K_3}\left(1 - 24\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n}\right),$$

$$\frac{d}{d\alpha}\frac{K_3'}{K_3} = -\frac{\sqrt{3}\pi\alpha'^3}{8\alpha^3 K_3^2},$$

$$\frac{dq}{d\alpha} = \frac{\pi^2\alpha'^3 q}{4\alpha^3 K_3^2}.$$

Proof. The derivative of α' with respect to α is straightforward since $\alpha' = \frac{\alpha}{\sqrt[3]{\alpha^3-1}}$. Now, we will find $\frac{dq}{d\alpha}$. We use theorem 2.2 and differentiate with respect to q to obtain

$$\frac{d\alpha}{dq} = 9\frac{d}{dq}q\prod_{n=1}^{\infty}\frac{\left(1-q^{9n}\right)^3}{\left(1-q^n\right)^3} = \frac{\alpha-1}{q}\left(1 + 3\sum_{n=1}^{\infty}\left(\frac{nq^n}{1-q^n} - \frac{9nq^{9n}}{1-q^{9n}}\right)\right).$$

We have that

$$1 + 3\sum_{n=1}^{\infty}\left(\frac{nq^n}{1-q^n} - \frac{9nq^{9n}}{1-q^{9n}}\right) = \frac{9P\left(q^9\right) - P\left(q\right)}{8},$$

where $P(q) = 1 - 24\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n}$. From corollary 4.6, we have

$$\frac{3P\left(q^3\right) - P\left(q\right)}{2} = \frac{12\alpha^2 K_3^2}{\pi^2}.$$

Using theorem 5.2, we get that

$$\frac{9P\left(q^9\right) - 3P\left(q^3\right)}{2} = \frac{4\left(\alpha+2\right)^2 K_3^2}{\pi^2}.$$

From which, it follows that $\frac{9P\left(q^3\right)-P(q)}{8} = \frac{4\left(\alpha^2+\alpha+1\right)K_3^2}{\pi^2}$. Thus, $\frac{d\alpha}{dq} = \frac{4\left(\alpha^3-1\right)K_3^2}{\pi^2 q} = \frac{4\alpha^3 K_3^2}{\pi^2\alpha'^3 q}$. Taking the reciprocal gives $\frac{dq}{d\alpha}$.

To prove the $\frac{dk_3}{d\alpha}$, we take the logarithmic derivative of the product formula for K_3 to obtain

$$\frac{dK_3}{d\alpha} = \frac{K_3}{q}\frac{dq}{d\alpha}\sum_{n=1}^{\infty}\left(\frac{3nq^{3n}}{1-q^{3n}} - \frac{3nq^n}{1-q^n}\right) = \frac{3\pi^2\alpha'^3}{4\alpha^3 K_3}\sum_{n=1}^{\infty}\left(\frac{nq^{3n}}{1-q^{3n}} - \frac{nq^n}{1-q^n}\right).$$

Using corollary 4.6, we get

$$\sum_{n=1}^{\infty}\frac{nq^{3n}}{1-q^{3n}} = -\frac{\alpha^2 K_3^2}{3\pi^2} + \frac{1}{36} + \frac{1}{3}\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n}.$$

We can substitute this into the expression for $\frac{dK_3}{d\alpha}$ to get the finalized expression for $\frac{dK_3}{d\alpha}$.

To prove $\frac{d}{d\alpha}\frac{K_3'}{K_3}$, we use $R = \frac{K_3'}{K_3}$ and $q = e^{-\frac{2\pi R}{\sqrt{3}}}$. We then have that $\frac{dq}{d\alpha} = -\frac{2\pi q}{\sqrt{3}}\frac{dR}{d\alpha} = \frac{\pi^2\alpha'^3 q}{4\alpha^3 K_3^2}$. We then solve for $\frac{dR}{d\alpha}$. □

This reparameterization enables us to perform calculus on α-K-formulas, efficiently producing both q-series and q-products.

Corollary 6.3. *Let* $q = e^{-\frac{2\pi R}{\sqrt{3}}} = e^{-\frac{2\pi}{\sqrt{3}}\frac{K_3'}{K_3}}$. *Then,*

$$\left(\frac{3\alpha'}{\alpha}\right)^{\frac{1}{4}} q^{\frac{1}{12}} = \prod_{n=0}^{\infty}\left(1-q^{3n+1}\right)\left(1-q^{3n+2}\right) = \prod_{n=1}^{\infty}\frac{1}{1+q^n+q^{2n}},$$

$$q^{-\frac{1}{24}}\left(\frac{3\alpha}{\alpha'}\right)^{\frac{1}{8}}\sqrt{\frac{2K_3}{\pi}} = \prod_{n=1}^{\infty}\left(1-q^n\right),$$

$$q^{-\frac{1}{8}}\left(\frac{\alpha^3}{3\alpha'^3}\right)^{\frac{1}{8}}\sqrt{\frac{2K_3}{\pi}} = \prod_{n=1}^{\infty}\left(1-q^{3n}\right).$$

Proof. We use corollary 4.6

$$\frac{12\alpha^2 K_3^2}{\pi^2 q} = \frac{1}{q} + 12\sum_{n=0}^{\infty}\left(\frac{(3n+1)\,q^{3n}}{1-q^{3n+1}} + \frac{(3n+2)\,q^{3n+1}}{1-q^{3n+2}}\right)$$

and integrate both sides with respect to q :

$$\int\frac{12\alpha^2 K_3^2}{\pi^2 q}dq = \log(q) - 12\log\left(\prod_{n=0}^{\infty}\left(1-q^{3n+1}\right)\left(1-q^{3n+2}\right)\right) + C_1.$$

However, we have that

$$\int\frac{12\alpha^2 K_3^2}{\pi^2 q}dq = \int\frac{3\alpha^2}{\alpha^3-1}d\alpha = \log\left(\alpha^3-1\right) = 3\log\left(\frac{\alpha}{\alpha'}\right).$$

Thus,

$$q^{\frac{1}{12}}\left(\frac{\alpha'}{\alpha}\right)^{\frac{1}{4}} = C\prod_{n=0}^{\infty}\left(1-q^{3n+1}\right)\left(1-q^{3n+2}\right).$$

Using the modular equation of degree 3 in theorem 5.2, we have

$$\left(\frac{q}{\alpha-1}\right)^{\frac{1}{4}}\left(9\left(\alpha^2+\alpha+1\right)\right)^{\frac{1}{12}}=C\prod_{n=0}^{\infty}\left(1-q^{3(3n+1)}\right)\left(1-q^{3(3n+2)}\right).$$

We let $\alpha\to 1$, so that $q\to 0$, to get

$$C=\lim_{\alpha\to 1}q^{\frac{1}{12}}\left(\frac{\alpha'}{\alpha}\right)^{\frac{1}{4}}=\lim_{\alpha\to 1}\left(\frac{q}{\alpha-1}\right)^{\frac{1}{4}}\left(9\left(\alpha^2+\alpha+1\right)\right)^{\frac{1}{12}}.$$

We have that $\lim_{\alpha\to 1}q^{\frac{1}{12}}\left(\frac{\alpha'}{\alpha}\right)^{\frac{1}{4}}=\left(\frac{1}{3}\lim_{\alpha\to 1}\frac{q}{\alpha-1}\right)^{\frac{1}{12}}$. Thus, if we let $x=\lim_{\alpha\to 1}\frac{q}{\alpha-1}$, we get $\left(\frac{x}{3}\right)^{\frac{1}{12}}=(3x)^{\frac{1}{4}}=C$. Therefore, $x=\frac{1}{9}$ and $C=3^{-\frac{1}{4}}$, which proves the first product formula.

To prove the second product formula, we note that

$$q^{\frac{1}{12}}\left(\frac{3\alpha'}{\alpha}\right)^{\frac{1}{4}}=\prod_{n=0}^{\infty}\left(1-q^{3n+1}\right)\left(1-q^{3n+2}\right)=\prod_{n=1}^{\infty}\frac{1-q^n}{1-q^{3n}}.$$

However, by definition of K_3, we have $\frac{2\sqrt{3}K_3}{\pi}=\prod_{n=1}^{\infty}\frac{(1-q^n)^3}{1-q^{3n}}$. Thus, dividing the two products, we get

$$q^{-\frac{1}{12}}\left(\frac{3\alpha}{\alpha'}\right)^{\frac{1}{4}}\frac{2K_3}{\pi}=\prod_{n=1}^{\infty}\left(1-q^n\right)^2.$$

We take square roots on both sides to prove the second product formula. To obtain the third product formula, we can just use the modular equation of degree 3 from theorem 5.2 on the α-K formula for $\prod_{n=1}^{\infty}\left(1-q^n\right)$. □

Using the modular equation of degree 2 in theorem 5.3, we can also find formulas for $\prod_{n=1}^{\infty}\left(1-q^{2n}\right)$ and $\prod_{n=1}^{\infty}\left(1+q^n\right)$.

In this framework, it is most natural to use $K_3=\frac{3}{4}\int_0^1\frac{dt}{y^2(t,\alpha)+\alpha t}$ as our definition of K_3, where $y=y(t,\alpha)$ satisfies $y^3+3\alpha ty+t^3=1$. However, this integral definition can be rather unwieldy. So we yearn to find a simpler expression for K_3 as a function of α. To do so, we will use theory of Ramanujan's elliptic functions of base 3, as developed by Berndt, Garvan, and Bhargava in [5], and relate K_3 to the hypergeometric function ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$.

Theorem 6.4. *Let $|\alpha|>1$. Then,*

$$K_3=\frac{\pi}{2\sqrt{3}\alpha}\,{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;\frac{1}{\alpha'^3}\right)=\frac{3}{4\alpha}\int_0^1\frac{dt}{(1-t^3)^{\frac{1}{3}}\left(1-\frac{t^3}{\alpha'^3}\right)^{\frac{2}{3}}},$$

$$K_3'=\frac{\pi}{2\sqrt{3}\alpha}\,{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;\frac{1}{\alpha^3}\right)=\frac{3}{4\alpha}\int_0^1\frac{dt}{(1-t^3)^{\frac{1}{3}}\left(1-\frac{t^3}{\alpha^3}\right)^{\frac{2}{3}}}.$$

Proof. We want to find a relationship between x and α such that $q_3 = \exp\left(-\frac{2\pi}{\sqrt{3}}\frac{K_3'(\alpha)}{K_3(\alpha)}\right) = \exp\left(-\frac{2\pi}{\sqrt{3}}\frac{{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;1-x\right)}{{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)}\right)$. Then, we have that

$$ {}_2F_1^2\left(\frac{1}{3},\frac{2}{3};1;x\right) = \frac{12\alpha^2K_3^2(\alpha)}{\pi^2} = 1+12\sum_{n=0}^{\infty}\left(\frac{(3n+1)\,q_3^{3n+1}}{1-q_3^{3n+1}} + \frac{(3n+2)\,q_3^{3n+2}}{1-q_3^{3n+2}}\right). $$

The equality between the series and ${}_2F_1^2\left(\frac{1}{3},\frac{2}{3};1;x\right)$ is provided by Berndt, Bhargava, and Garvin [5]. The equivalence between $\frac{12\alpha^2K_3^2}{\pi^2}$ and the series is from corollary 4.6. Berndt, Bhargava, and Garvan also provided a modular equation of degree 3 for Ramanujan's elliptic integral ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$. Using the modular equation of degree 3 for ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$ in [5] and theorem 5.2, we get

$$ \frac{\left(1+2\,(1-x)^{\frac{1}{3}}\right)^2}{9}\ {}_2F_1^2\left(\frac{1}{3},\frac{2}{3};1;x\right) = \frac{4\,(\alpha+2)^2\,K_3^2(\alpha)}{3\pi^2} $$

$$ = 1+12\sum_{n=0}^{\infty}\left(\frac{(3n+1)\,q_3^{3(3n+1)}}{1-q_3^{3(3n+1)}} + \frac{(3n+2)\,q_3^{3(3n+2)}}{1-q_3^{3(3n+2)}}\right). $$

Thus, we have the following system of equations:

$$ \frac{\left(1+2\,(1-x)^{\frac{1}{3}}\right)^2}{9}\ {}_2F_1^2\left(\frac{1}{3},\frac{2}{3};1;x\right) = \frac{4\,(\alpha+2)^2\,K_3^2}{3\pi^2}, $$

$$ {}_2F_1^2\left(\frac{1}{3},\frac{2}{3};1;x\right) = \frac{12\alpha^2K_3^2(\alpha)}{\pi^2}. $$

We can solve the system for x and ${}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$ to obtain theorem 6.3. □

7. Connection with Jacobi's Elliptic Integral

The connection between the elliptic integral K_3 and Jacobi's first complete elliptic integral, which will be denoted as K_2, can be stated by the following relationship between hypergeometric functions:

Theorem 7.1.

$$ {}_2F_1\left(\frac{1}{3},\frac{2}{3};1;\frac{27x\,(1-x)^4\,(1-2x)}{2\,(1-3x^2)^3}\right) = \frac{1-3x^2}{\sqrt{1-2x}}\ {}_2F_1\left(\frac{1}{2},\frac{1}{2};1;\frac{(2-3x)^3\,x}{1-2x}\right), $$

$$ \frac{2}{\sqrt{3}}\ \frac{{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;\frac{(2-3x)(1-3x)^4}{2(1-3x^2)^3}\right)}{{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;\frac{27x(1-x)^4(1-2x)}{2(1-3x^2)^3}\right)} = \frac{{}_2F_1\left(\frac{1}{2},\frac{1}{2};1;\frac{(1-x)(1-3x)^3}{1-2x}\right)}{{}_2F_1\left(\frac{1}{2},\frac{1}{2};1;\frac{(2-3x)^3x}{(1-2x)}\right)}. $$

Proof. Let $K_3 = K_3(\alpha) = \frac{3}{4\alpha}\int_0^1\left(1-t^3\right)^{-\frac{1}{3}}\left(1-\frac{t^3}{\alpha'^3}\right)^{-\frac{2}{3}}dt$, $K_3' = \frac{\alpha'}{\alpha}K_3(\alpha')$, $\alpha' = \frac{\alpha}{\sqrt[3]{\alpha^3-1}}$, and $q_3 = \exp\left(-\frac{2\pi}{\sqrt{3}}\ \frac{K_3'}{K_3}\right)$.

Let $K_2 = K_2(k) = \int_0^1\frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}$, $K_2' = K_2(k')$, $k' = \sqrt{1-k^2}$,

$q_2 = \exp\left(-\pi \frac{K_2'}{K_2}\right)$. We seek a relationship between α and k by imposing the restriction that $\frac{2}{\sqrt{3}} \frac{K_3'(\alpha)}{K_3(\alpha)} = \frac{K_2'(k)}{K_2(k)}$, so that $q_3 = q_2 = q$. It is known that

$$\prod_{n=1}^{\infty} (1 - q^n) = 2^{\frac{1}{3}} q^{-\frac{1}{24}} k^{\frac{1}{12}} k'^{\frac{1}{3}} \sqrt{\frac{K_2}{\pi}},$$

$$\prod_{n=1}^{\infty} \left(1 - q^{2n}\right) = q^{-\frac{1}{12}} \left(2kk'\right)^{\frac{1}{6}} \sqrt{\frac{K_2}{\pi}}.$$

Using corollary 6.3, we have

$$\prod_{n=1}^{\infty} (1 - q^n) = q^{-\frac{1}{24}} \left(\frac{3\alpha}{\alpha'}\right)^{\frac{1}{8}} \sqrt{\frac{2K_3}{\pi}}.$$

Using theorem 5.3, we also have that

$$\prod_{n=1}^{\infty} \left(1 - q^{2n}\right) = q^{-\frac{1}{12}} \left(g^3 - 8\right)^{\frac{1}{12}} \left(g^3 + 1\right)^{\frac{1}{24}} g^{\frac{1}{4}} \sqrt{\frac{K_3}{\pi}},$$

where $g = 2\sqrt{\alpha} \cos\left(\frac{\arctan \sqrt{\alpha^3 - 1}}{3}\right) = \sqrt[3]{1 + \sqrt{1 - \alpha^3}} + \sqrt[3]{1 - \sqrt{1 - \alpha^3}}$. This gives us a system of equation between α, k, K_3, and K_2. Let $b = \frac{8}{g^4} \sqrt{\frac{g^3+1}{g(g^3-8)}}$, then we have that

$$k = \sqrt{1 + b^2} - b,$$

$$k' = \sqrt{2b\left[\sqrt{1 + b^2} - b\right]},$$

$$K_3(\alpha) = 2^{-\frac{1}{3}} \left(\frac{\alpha'}{3\alpha}\right)^{\frac{1}{4}} k^{\frac{1}{6}} k'^{\frac{2}{3}} K_2(k),$$

$$\frac{2}{\sqrt{3}} \frac{K_3'(\alpha)}{K_3(\alpha)} = \frac{K_2'(k)}{K_2(k)}.$$

We can write the equations above in the following form:

$$K_3\left(\frac{\sqrt[3]{2}\left(1 - 3x^2\right)}{(2 - 3x)^{\frac{1}{3}} (1 - 3x)^{\frac{4}{3}}}\right) = \frac{(2 - 3x)^{\frac{1}{3}} (1 - 3x)^{\frac{4}{3}}}{2^{\frac{1}{3}} 3^{\frac{1}{2}} (1 - 2x)^{\frac{1}{2}}} K_2\left(\frac{(2 - 3x)^{\frac{3}{2}} x^{\frac{1}{2}}}{(1 - 2x)^{\frac{1}{2}}}\right),$$

$$\frac{2}{\sqrt{3}} \frac{K_3'\left(\frac{\sqrt[3]{2}\left(1-3x^2\right)}{(2-3x)^{\frac{1}{3}}(1-3x)^{\frac{4}{3}}}\right)}{K_3\left(\frac{\sqrt[3]{2}(1-3x^2)}{(2-3x)^{\frac{1}{3}}(1-3x)^{\frac{4}{3}}}\right)} = \frac{K_2'\left(\frac{(2-3x)^{\frac{3}{2}} x^{\frac{1}{2}}}{(1-2x)^{\frac{1}{2}}}\right)}{K_2\left(\frac{(2-3x)^{\frac{3}{2}} x^{\frac{1}{2}}}{(1-2x)^{\frac{1}{2}}}\right)},$$

for $0 \leq x < \frac{1}{3}$. We the use the hypergeometric function representations of K_3 and K_2 to finish the proof. $\square$

This connection allows us to find some singular moduli for K_2. For example, if we put $x = 2-\sqrt{3}+\sqrt{2}-2\sqrt{\frac{2}{3}}$ in theorem 7.1, we get

$$K_3\left(\sqrt[3]{2}\right) = \frac{2^{\frac{2}{3}}}{3^{\frac{1}{2}}}\sqrt{-93-66\sqrt{2}+54\sqrt{3}+38\sqrt{6}}K_2\left(15-6\sqrt{6}+10\sqrt{2}-8\sqrt{3}\right) = \frac{\sqrt{3}\Gamma^3\left(\frac{1}{3}\right)}{2^{\frac{11}{3}}\pi}.$$

Furthermore, we have that $\frac{K_3'\left(\sqrt[3]{2}\right)}{K_3\left(\sqrt[3]{2}\right)} = 1$, so it follows that

$$\frac{K_2'\left(15-6\sqrt{6}+10\sqrt{2}-8\sqrt{3}\right)}{K_2\left(15-6\sqrt{6}+10\sqrt{2}-8\sqrt{3}\right)} = \frac{2}{\sqrt{3}},$$

$$K_2\left(15-6\sqrt{6}+10\sqrt{2}-8\sqrt{3}\right) = \frac{1}{\sqrt{-93-66\sqrt{2}+54\sqrt{3}+38\sqrt{6}}}\frac{3\sqrt[3]{4}\Gamma^3\left(\frac{1}{3}\right)}{32\pi}.$$

Through this link, we see that modular equations for K_3 imply singular moduli for K_2 such that $\frac{K_2'}{K_2} = n\sqrt{3}$ for n a rational number.

8. Ramanujan's series for $\frac{1}{\pi}$

Lastly, we will conclude this paper by giving a few more series for $\frac{1}{\pi}$ involving the hypergeometric series ${}_3F_2\left(\frac{1}{3},\frac{2}{3},\frac{1}{2};1,1;x\right)$. In his 1914 paper "Modular Equations and Approximations to π", Ramanujan gave two series for this signature. We will provide a few more using this theory.

Theorem 8.1. *Let* $j_3 = (3\alpha\alpha')^3 = \frac{1}{q}+42+783q+8672q^2+...$, $P(q) = 1-24\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n}$, $A_3\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right) = \left(P\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)-\frac{3\sqrt{3}}{\pi\ Re(R)}\right)\frac{\pi^2}{K_3^2}$, *and* $C_3 = \frac{48-\alpha A_3\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)-36\alpha^3}{72(2-\alpha^3)}$. *Then, for* $|j_3|>108$,

$$\frac{1}{\pi} = 2\Re(R)\sqrt{\frac{1}{3}\left(1-\frac{108}{j_3}\right)}\sum_{n=0}^{\infty}\frac{(C_3+n)\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{108}{j_3}\right)^n.$$

Here, K_3 is best thought of as a function of R via its product definition.

Proof. We start by noting that ${}_2F_1\left(\frac{1}{6},\frac{1}{3};1;4x(1-x)\right) = {}_2F_1\left(\frac{1}{3},\frac{2}{3};1;x\right)$. Thus, using theorem 6.4, corollary 4.6, and Clausen's formula, we get

$$\frac{12\alpha^2K_3^2}{\pi^2} = \sum_{n=0}^{\infty}\frac{\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{4}{\alpha^3\alpha'^3}\right)^n.$$

Logarithmically differentiating the equation above and using theorem 6.2, we get

$$\frac{2\alpha^5\left(4-\alpha'^3\right)}{\alpha'^3-\alpha^3}\frac{K_3^2}{\pi^2}+\frac{\alpha^3\alpha'^3}{6\left(\alpha'^3-\alpha^3\right)}P(q) = \sum_{n=0}^{\infty}\frac{n\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{4}{\alpha^3\alpha'^3}\right)^n,$$

where $P(q) = 1-24\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n}$ and $q = \exp\left(-\frac{2\pi}{\sqrt{3}}\frac{K_3'}{K_3}\right)$. These two equations are key for Ramanujan's series for π. The idea is that if $q = \exp\left(-\frac{2\pi R}{\sqrt{3}}\right)$ for R a positive rational number or quadratic irrational with positive real part, then $P(q)$ will be of the form $A_3\frac{K_3^2}{\pi^2} + \frac{3\sqrt{3}}{\pi\Re(R)}$, where A_3 is an algebraic number. This is also true if we have $-q$ instead of q. Thus, we can combine the two series in such a way so that we eliminate the $\frac{K_3^2}{\pi^2}$ term, giving us a series for $\frac{1}{\pi}$.

In general, we write $P\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right) = A_3\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)\frac{K_3^2}{\pi^2} + \frac{3\sqrt{3}}{\pi\ Re(R)}$, so that we get the following two equations:

$$\frac{12\alpha^2K_3^2}{\pi^2} = \sum_{n=0}^{\infty}\frac{\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{4}{\alpha^3\alpha'^3}\right)^n,$$

$$-\left[\frac{48-\alpha A_3\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)-36\alpha^3}{72\left(2-\alpha^3\right)}\right]\frac{12\alpha^2K_3^2}{\pi^2}+\frac{\alpha^3\alpha'^3}{\alpha'^3-\alpha^3}\frac{\sqrt{3}}{2\pi\ \Re\left(R\right)} = \sum_{n=0}^{\infty}\frac{n\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{4}{\alpha^3\alpha'^3}\right)^n.$$

Since $j_3 = (3\alpha\alpha')^3$, we have that $\frac{4}{\alpha^3\alpha'^3} = \frac{108}{j_3}$ and $\frac{\alpha^3\alpha'^3}{\alpha'^3-\alpha^3} = \frac{1}{\sqrt{1-\frac{108}{j_3}}}$. We then put $C_3 = \frac{48-\alpha A_3\left(e^{-\frac{2\pi R}{\sqrt{3}}}\right)-36\alpha^3}{72(2-\alpha^3)}$. This lets us rewrite the sums as

$$\sum_{n=0}^{\infty}\frac{\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{108}{j_3}\right)^n = \frac{12\alpha^2K_3^2}{\pi^2},$$

$$\sum_{n=0}^{\infty}\frac{n\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{108}{j_3}\right)^n = -C_3\frac{12\alpha^2K_3^2}{\pi^2}+\frac{1}{\sqrt{1-\frac{108}{j_3}}}\frac{\sqrt{3}}{2\pi\ \Re\left(R\right)},$$

so that

$$\sum_{n=0}^{\infty}\frac{\left(C_3+n\right)\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n\left(\frac{1}{2}\right)_n}{(n!)^3}\left(\frac{108}{j_3}\right)^n = \frac{1}{\sqrt{1-\frac{108}{j_3}}}\frac{\sqrt{3}}{2\pi\ \Re\left(R\right)}.$$

We then solve for $\frac{1}{\pi}$ to get the series formula. We note that the series only converges if $|j_3| > 108$. □

It should be noted that j_3 is analogous to the j-invariant in this theory, as it is invariant under $R \to R + i\sqrt{3}$ and $R \to \frac{1}{R}$. The function C_3 satisfies the same transformations, but is not holomorphic anywhere. The function C_3 is a non-meromorphic modular function of weight 0.

Theorem 8.2. *Let $q = e^{-\frac{2\pi R}{\sqrt{3}}}$, where R is positive rational or a quadratic irrational with positive real part. Then, the only integral values of j_3 are -27000000, -110592, -326592, -8640, -1728, -192, -27, 0, 8, 64, 108, 216, 1458, and 3375. These values of j_3 occur when $q = -e^{-\pi\sqrt{\frac{89}{3}}}$,*

$-e^{-\pi\sqrt{\frac{41}{3}}}$, $-e^{-\frac{7\pi}{\sqrt{3}}}$, $-e^{-\frac{5\pi}{\sqrt{3}}}$, $-e^{-\pi\sqrt{\frac{17}{3}}}$, $-e^{-\pi\sqrt{3}}$, $-e^{-\pi\sqrt{\frac{5}{3}}}$, $-e^{-\frac{\pi}{\sqrt{3}}}$, $\omega^2 e^{-\frac{2\sqrt{2}\pi}{3}}$, $-\omega^2 e^{-\frac{\pi\sqrt{11}}{3}}$, $e^{-\frac{2\pi}{\sqrt{3}}}$, $e^{-2\pi\sqrt{\frac{2}{3}}}$, $e^{-\frac{4\pi}{\sqrt{3}}}$, *and* $e^{-2\pi\sqrt{\frac{5}{3}}}$ *respectively.*

Proof. When $q = e^{-\frac{2\pi R}{\sqrt{3}}} = e^{-\frac{2\pi}{\sqrt{3}}\frac{K_3'}{K_3}}$, we have that $j_3 = (3\alpha\alpha')^3$. Using theorem 4.2, we can see that $j = \frac{27\alpha^3\left(9\alpha^3-8\right)^3}{\alpha^3-1}$, where j is the usual j-invariant. Thus, we can see that j and j_3 satisfy the equation $j^2 - j_3\left(j_3^2 - 126j_3 + 2944\right)j + j_3\left(j_3+192\right)^3 = 0$. This equation is quadratic in j. Thus, whenever j_3 is an integer and $q = e^{-\frac{2\pi R}{\sqrt{3}}}$ with R a positive rational or quadratic irrational with positive real part, j is either a rational (more precisely, an integer) or a quadratic rational. This corresponds to the class 1 and class 2 values of j, and there are only finitely many such values. Furthermore, these values have been well-studied and all have been classified along with the corresponding value of q. For each class 1 and class 2 value of j, we can solve for j_3 and note the ones that result in j_3 being an integer as well as the corresponding integer value of j_3. Doing so, gives the list in theorem 8.2.

□

Thus, we can see that certain values of q yield elegant series for $\frac{1}{\pi}$, particularly the values where j_3 is an integer such that $|j_3| > 108$. The value of C_3 is rational for integral j_3, so it is only a matter of computation to find the rational number matching C_3. Using C_3, j_3, q and its associated R value, we can use theorem 8.1 to get a series for $\frac{1}{\pi}$. Here is a table listing all such series:

q	j_3	Series for $\frac{1}{\pi}$
$-e^{-\pi\sqrt{\frac{89}{3}}}$	-27000000	$\frac{1}{\pi} = \frac{1}{3\sqrt{3}}\sum_{n=0}^{\infty}\frac{(-1)^n(14151n+827)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\frac{1}{500^{2n+1}}$
$-e^{-\frac{7\pi}{\sqrt{3}}}$	-326592	$\frac{1}{\pi} = \frac{\sqrt{7}}{108}\sum_{n=0}^{\infty}\frac{(-1)^n(165n+13)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\frac{1}{3024^n}$

q	j_3	Series for $\frac{1}{\pi}$
$-e^{-\pi\sqrt{\frac{41}{3}}}$	-110592	$\frac{1}{\pi} = \frac{\sqrt{3}}{288}\sum_{n=0}^{\infty}\frac{(-1)^n(615n+53)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\frac{1}{2^{10n}}$
$-e^{-\frac{5\pi}{\sqrt{3}}}$	-8640	$\frac{1}{\pi} = \frac{\sqrt{15}}{12}\sum_{n=0}^{\infty}\frac{(-1)^n(9n+1)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\frac{1}{80^n}$
$-e^{-\pi\sqrt{\frac{17}{3}}}$	-1728	$\frac{1}{\pi} = \frac{1}{3\sqrt{3}}\sum_{n=0}^{\infty}\frac{(-1)^n(51n+7)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\frac{1}{4^{2n+1}}$
$-e^{-\pi\sqrt{3}}$	-192	$\frac{1}{\pi} = \frac{1}{\sqrt{3}}\sum_{n=0}^{\infty}\frac{(-1)^n(5n+1)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\left(\frac{3}{4}\right)^{2n+1}$
$e^{-2\pi\sqrt{\frac{2}{3}}}$	216	$\frac{1}{\pi} = \frac{1}{3\sqrt{3}}\sum_{n=0}^{\infty}\frac{(6n+1)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\frac{1}{2^n}$
$e^{-\frac{4\pi}{\sqrt{3}}}$	1458	$\frac{1}{\pi} = 2\sum_{n=0}^{\infty}\frac{(15n+2)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\left(\frac{2}{27}\right)^{n+1}$
$e^{-2\pi\sqrt{\frac{5}{3}}}$	3375	$\frac{1}{\pi} = \frac{2}{15\sqrt{3}}\sum_{n=0}^{\infty}\frac{(33n+4)(\frac{1}{3})_n(\frac{2}{3})_n(\frac{1}{2})_n}{(n!)^3}\left(\frac{4}{125}\right)^{n}$

It can be seen that the first series converges the most rapidly among all the other series above. We can also rewrite the last series in the form

$$\frac{1}{\pi} = 20\sqrt{3}\sum_{n=0}^{\infty}\frac{(-1)^n\,(2n)!\,(3n)!\,(827+14151n)}{(n!)^5\,300^{3n+2}}.$$

All of the series shown in the table above were given by Chang, Liaw, and Tan in [8]. Their approach relied on heavy computational machinery by computing modular equations of various degrees. The last two series were given by Ramanujan in [9]. If we allow j_3 to be non-integral (e.g. quadratic irrational), we can find more series for $\frac{1}{\pi}$.

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